

Chapter 10: Google Search Operators

Syntax, Structure, and Disciplined Query Construction

Companion Workbook to the Instructional Video

How to Use This Workbook

This is the written companion to the instructional video on Google search operators. It works as a follow-along guide, a standalone textbook chapter, and a practice manual.

Where this chapter sits in the series. Chapter 9 covered search engines as a class: how they crawl, index, rank, and retrieve, and why no single index is the web. **This chapter goes one level deeper into one engine's query language**, because Google has the richest and most reliable operator support, and because operator skill is the difference between searching and finding.

Everything Chapter 9 established still applies. Operators filter and reorder what is already in the index; **no operator can retrieve a page the crawler never fetched**. Rule 10 governs every null result you produce in this chapter.

A warning about volatility. Operator behaviour changes without notice, and several once-standard operators have been withdrawn. Some operators are officially documented; many are not, and undocumented operators are supported at Google's discretion and can stop working at any time.

Section 1.8 covers what has already been withdrawn, and Section 1.1.5 gives you a method for testing whether an operator still works. Treat every syntax table in this chapter as a snapshot to verify rather than a specification.

Time estimate:

Component	Estimated time
Video	60 to 75 minutes
Reading material	4 to 5 hours
Knowledge assessment	1 hour
Critical thinking exercises	2 hours
Labs 1 through 5	11 to 15 hours
Capstone assignment	5 to 8 hours
Total	24 to 32 hours

Materials: a computer with internet access and a browser. All labs involve live searching except Lab 5, which is entirely offline, and Lab 4, which has an offline path.

The Rules Still Apply

Rule 1: No private individuals. Rule 2: Publicly available means genuinely public. Rule 3: Document as you go. Rule 4: No dark web access. Rule 5: Automation is constrained. Rule 6: A relationship is a claim, and every claim needs a basis. Rule 7: If it is not written down, it did not happen. Rule 8: Lawful is the floor, not the ceiling. Rule 9: OPSEC never buys access. Rule 10: Absence of results is a statement about the index, not about the world.

This chapter adds the rule that governs operator use specifically:

Rule 11: An operator that finds something does not authorize accessing it.

Search operators are extremely good at surfacing material that was published by accident: configuration files, backups, credential stores, internal documents, exposed interfaces, and directory listings. Collections of such queries circulate widely.

Chapter 7's position applies without modification. Material exposed by error was not published, and availability by mistake is not publication. Finding a login page does not authorize logging in. Finding an exposed file does not authorize downloading it. Finding an exposed system does not authorize touching it in any way. The correct response to an accidental exposure is Chapter 7's coordinated disclosure process, not collection.

In this course: you will learn the operators, including the ones that surface exposures, because understanding them is necessary for defensive work and for auditing your own organization. You will not use them against systems you do not own or are not authorized to test. Lab 3 is a website audit performed with **read-only public queries against a site's indexed content**, which is a legitimate and useful exercise, and it explicitly forbids any attempt to reach restricted resources.

Rule 5 applies with particular force here. Automated querying is prohibited by Google's terms, is detected routinely, and results in blocking. Every lab is performed manually.

Learning Objectives

Knowledge objectives (you can explain it):

1. Explain what an operator does mechanically, and why no operator expands the index.
2. Explain Boolean logic as Google implements it, including implicit AND, OR, exclusion, and grouping.
3. Describe each operator covered in this chapter: its syntax, its purpose, its behaviour, and its limitations.
4. Distinguish the `in`-prefixed operators from the `all`-prefixed forms and explain why the difference matters.
5. Explain how operators combine, which combinations conflict, and why order sometimes matters.

6. Explain the difference between a documented and an undocumented operator, and the risk each carries.
7. Identify the operators that have been withdrawn and explain what replaced them, where anything did.
8. Explain the limits of operators: index dependence, silent failure, approximation, and result depth restriction.
9. Explain the ethical and legal boundary around operators that surface accidental exposures.

Application objectives (you can do it):

1. Construct precise queries using exact phrase, exclusion, OR, grouping, and wildcards.
2. Use structural operators to target a domain, a URL pattern, a title, or body text.
3. Use file type and source filters to find documents rather than pages.
4. Use temporal and proximity operators to control recency and word relationships.
5. Combine four or more operators into a single effective query.
6. Diagnose a failing query and repair it.
7. Test whether an operator is working, rather than assuming it.
8. Document queries to the Chapter 6 and Chapter 9 standard.

Analysis objectives (you can judge it):

1. Evaluate whether a query's results reflect the query or the index.
2. Judge when an operator is the wrong tool and a structured source is required.
3. Assess a query's precision and recall trade-off and adjust deliberately.
4. Recognize when an operator has failed silently.

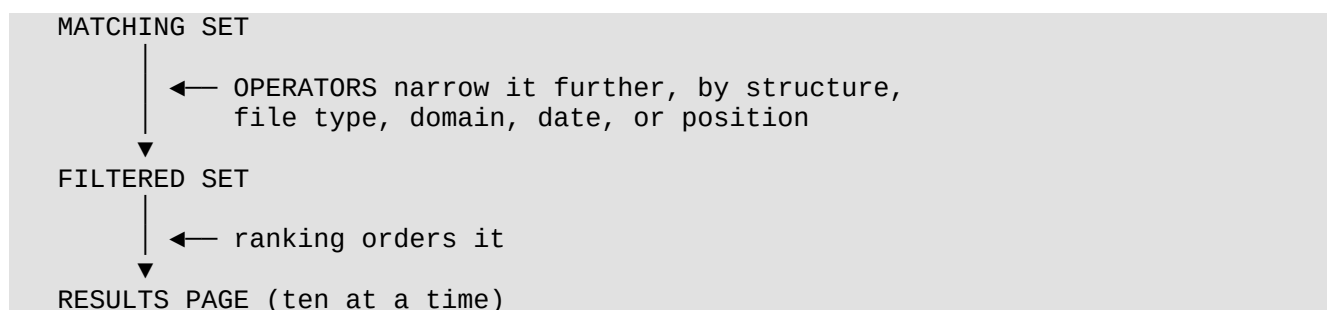
Part 1: Reading Material

1.1 Introduction to Google Search Operators

1.1.1 What an Operator Is

A **search operator** is a special instruction in a query that constrains or modifies how the engine matches and returns results. Operators are not magic and they do exactly one kind of thing: **they filter and reorder what is already in the index.**

THE INDEX (everything Google has crawled and kept)
↓
← your query terms narrow this to matching pages
▼



Two consequences follow, and both are on the exam.

Operators cannot add to the index. A perfectly constructed query cannot return a page Google never crawled, never indexed, or has dropped. This is Chapter 9's Rule 10 restated at the query level.

Operators change precision, not existence. A well-constructed query finds the same material faster and with less noise. A badly constructed one buries it. Neither creates it.

1.1.2 Why Operators Matter

Without operators	With operators
"annual report riverside manufacturing" returns news, aggregators, and directory listings	A file type restriction plus a domain restriction returns the document itself
A common word dominates every result	Exclusion removes the dominant meaning in one character
A phrase is matched loosely and out of order	Quotation marks require the exact string
Recent content buries the original source	Date restriction inverts the ordering problem
An organization's own material is scattered through thousands of results	A site restriction returns only their pages

The measurable effect is usually an order of magnitude: a query returning 40,000 results becomes one returning 30, with the useful items on the first page. That is the entire value proposition, and it is substantial.

1.1.3 Precision and Recall

Two competing goals, and every operator choice trades between them.

	Precision	Recall
Definition	The proportion of returned results that are relevant	The proportion of relevant material that was returned
High when	Your query is tightly constrained	Your query is loose
Operators that raise it	Exact phrase, site restriction, file type, title restriction, exclusion	OR, wildcards, dropping restrictions
The cost	You may exclude relevant	You must read through noise

	Precision	Recall
	material you did not anticipate	
Investigative use	When you know what you are looking for	When you are establishing what exists

The working pattern: start with high recall to see the shape of the material, then raise precision to work it. Investigators who start narrow and stay narrow miss what they did not think to ask for; investigators who never narrow down.

1.1.4 Documented and Undocumented Operators

Google publishes documentation for a small subset of its operators. Many operators in common use are undocumented, which has practical consequences:

	Documented	Undocumented
Support commitment	Stated	None
Stability	Higher	Can change or vanish without notice
Behaviour	Described	Inferred from observation, sometimes wrongly
Examples of the type	Site restriction, file type, exact phrase, exclusion, before and after	Proximity, some structural operators, several specialized commands
Risk to your work	Low	A query that worked last month may silently behave differently

This is why Section 1.1.5 exists, and why professional practice tests operators rather than trusting them.

1.1.5 Testing Whether an Operator Works

The single most useful habit in this chapter, and it takes two minutes.

1. CHOOSE A QUERY WITH A KNOWN ANSWER
Something you can verify independently: a page you know exists, on a domain you know, with a title you know.
↓
2. RUN IT WITHOUT THE OPERATOR
Record the result count and whether your known page appears.
↓
3. RUN IT WITH THE OPERATOR
Record the same.
↓

- ▼
4. COMPARE
Did the result set change in the way the operator claims? Did your known page survive a filter it should have survived, and vanish from one it should not have?
↓
 5. TEST THE NEGATIVE
Construct a query the operator should return NOTHING for. If it returns results, the operator is being ignored.
↓
 6. RECORD THE RESULT AND THE DATE
Operator behaviour changes. Your test is a snapshot.

Step 5 is the one that catches silent failure. An operator that is being ignored produces results that look filtered because you expected them to be filtered, and the only reliable way to detect it is to construct a query that must return nothing if the operator is working.

Worked example. To test a file type restriction: search a common term restricted to a file type that is exceptionally rare on the web. If the operator works, you get few or no results. If you get millions of ordinary web pages, the operator is being ignored.

1.1.6 Documentation Discipline

Chapter 6 and Chapter 9 apply. Every query you rely on is logged with:

Field	Why
Exact query string, character for character	Operators are whitespace and case sensitive in places
Which operators it uses	So you can diagnose failures
Whether each operator appeared to work	Silent failure is common
Locale and language settings	Results vary substantially
Signed-in state	Personalization
Date, time, time zone	Indexes change
Reported result count	An approximation, and still a data point
Results page screenshot	Searches are not reproducible

A specific warning for this chapter. Copying a query out of a document frequently breaks it, because word processors substitute typographic quotation marks for straight ones and Google may not interpret them as an exactPhrase operator. When a query mysteriously stops working, check the quotation marks first.

1.2 Fundamentals and Boolean Logic

1.2.1 Implicit AND

Google treats multiple terms as requiring all of them, by default. Typing three words asks for pages containing all three, in any order and any position.

Query form	Behaviour
term1 term2 term3	All three required; order and adjacency not required
AND typed explicitly	Redundant; it is already the default

The practical consequence: adding a word narrows. Removing a word widens. This is the cheapest precision control available and it is used constantly without being noticed.

Stop words and short terms. Very common words are handled inconsistently and may be ignored in an unquoted query. If a common word matters, quote the phrase containing it.

1.2.2 Exact Phrase: Quotation Marks

Placing terms in straight double quotation marks requires them in that order, adjacent.

Query form	Meaning
"exact phrase here"	Those words, in that order
"phrase one" "phrase two"	Both phrases required, independently
word "phrase"	The word anywhere, plus the phrase

Why it is the highest-yield operator in investigative work:

Use	Example type
Names of organizations, documents, and products	An exact institutional name that is otherwise a common word combination
Document titles and headings	Finding a specific report
Error messages and technical strings	Diagnosing a problem; finding others who hit it
Boilerplate and template language	Provenance tracing; finding copied text (Chapter 5)
Job titles and formal role names	Corporate structure research
Quoted sentences	Establishing where a claim first appeared

Behaviour to know. Quotation marks also suppress some of Google's automatic query expansion, meaning synonyms, stemming, and spelling correction are reduced. This makes results more literal and can reduce recall sharply, which is the intended trade.

The **typographic quotation mark problem** from Section 1.1.6 applies most acutely here.

1.2.3 Exclusion: The Minus Sign

A hyphen immediately before a term, with no space, removes results containing that term.

Query form	Meaning
term -unwanted	Contains term, does not contain unwanted
term -"unwanted phrase"	Excludes the phrase
term -site:example.com	Excludes a whole domain
term -filetype:pdf	Excludes a file type

Exclusion works with most operators, which is what makes it powerful: you can exclude a domain, a file type, a title pattern, or a URL pattern, not just a word.

The classic uses:

Problem	Exclusion
A dominant unrelated meaning swamps results	Exclude a word characteristic of the unwanted meaning
A subject's own site dominates	Exclude their domain to find third-party coverage
Aggregators and job boards flood results	Exclude the dominant aggregator domains
A product name matches a celebrity or a film	Exclude terms from that context

The risk: over-exclusion removes relevant material silently. If a query suddenly returns very little, remove exclusions one at a time and observe.

1.2.4 OR and Grouping

OR in capitals, or the pipe character, requires either term.

Query form	Meaning
term1 OR term2	Either
term1 \ term2	Either; the pipe is equivalent
(term1 OR term2) term3	Either of the first two, and the third
"phrase one" OR "phrase two"	Either phrase

Case matters. Lowercase or is treated as an ordinary word and is usually ignored. This is one of the most common syntax errors and it fails silently, because the query still returns results, just the wrong ones.

Grouping with parentheses controls how OR combines with the implicit AND. Without parentheses,

the scope of an OR can be ambiguous, and Google's interpretation may not be yours.

The highest-value use of OR in investigative work is covering variant terminology in one query:

Situation	Query pattern
An entity with name variants	("Full Legal Name" OR "Trading Name" OR "Abbreviation")
A concept with several standard terms	(fine OR penalty OR "civil money penalty" OR sanction)
A document type with several names	("annual report" OR "yearly report" OR "annual review")
A role with several titles	("chief executive" OR "managing director" OR "president")

1.2.5 Wildcards

An asterisk acts as a placeholder for one or more words, most reliably inside a quoted phrase.

Query form	Meaning
"the * of the report"	The phrase with any word or words in that position
"appointed * as chief executive"	Finding the names filling a role, across many pages

Uses: finding phrase variants, discovering the words that fill a slot, and matching a sentence pattern whose middle you do not know.

Limitations: behaviour is looser than users expect, it does not work as a partial-word wildcard, and it is unreliable outside quoted phrases.

1.2.6 Numeric Ranges

Two dots between numbers match any value in the range.

Query form	Meaning
report 2015..2020	Any number in that range appearing on the page
"annual report" 2018..2024 site:example.org	Combined with other operators
capacity 500..2000 tonnes	Quantities

Uses: year ranges in document titles, price and capacity ranges, and any question where a value falls within a band. Behaviour is approximate and it matches numbers as text on the page, not as structured data.

1.2.7 Boolean Discipline

Principle	Reason
Capitalize OR, always	Lowercase fails silently
Group anything ambiguous	Your reading of precedence and Google's may differ
Quote phrases containing common words	Otherwise the common word may be ignored
Use straight quotation marks	Typographic quotes may not register
Add exclusions one at a time	So you can see what each removes
Do not type AND	It is redundant and it clutters
Test complex Boolean structures in pieces	Build up, do not debug a five-operator query blind

1.3 Location and Structure Operators

These operators constrain **where in a page or a site** a term must appear. They are the operators that turn a search into a targeted retrieval.

1.3.1 Site Restriction

Operator	Form	Meaning
site:	site:example.com term	Only pages on that domain
	site:sub.example.com	Only that subdomain
	site:example.com/reports/	Only that path, where supported
	site:.gov	Only that top-level domain
	site:.ac.uk	Only that domain suffix
	-site:example.com	Excluding that domain

This is the most useful single operator in the chapter, for four reasons:

Enumerating an organization's own material. A site restriction with a broad term returns what an organization publishes, which is Chapter 3's point about job postings and published documents applied at scale.

Sector sweeps. Restricting to a country's government suffix, an academic suffix, or a sector-specific domain finds institutional material without knowing which institution.

Auditing. Comparing what a site restriction returns against what an organization believes it publishes is the basis of Lab 3.

Exclusion. Removing a dominant domain from results, which is frequently the fastest way to find

third-party coverage of an organization.

Behaviour notes. Path-level restriction support has varied over time; test it. Results returned by a site restriction are an approximation of what is indexed, not a complete inventory, and the reported count is unreliable for anything but rough comparison.

1.3.2 URL Operators

Operator	Form	Meaning
<code>inurl:</code>	<code>inurl:annual-report</code>	The term appears in the page URL
<code>allinurl:</code>	<code>allinurl:annual report 2023</code>	All following terms in the URL

Uses:

Purpose	Pattern
Finding a section of a site	A path element characteristic of that section
Finding document repositories	Path words like reports, publications, downloads, archive
Finding pages with a functional purpose	Path words characteristic of that function
Finding directory listings	Combined with title operators, per Section 1.3.5

Caution. `allinurl:` combines poorly with other operators; when in doubt, use repeated `inurl:` terms instead. And URL patterns that look like they identify sensitive functionality are exactly where Rule 11 applies.

1.3.3 Title Operators

Operator	Form	Meaning
<code>intitle:</code>	<code>intitle:"annual report"</code>	The phrase appears in the page title
<code>allintitle:</code>	<code>allintitle:annual report 2023</code>	All following terms in the title

Why titles matter. A page's title is the closest thing to a machine-readable statement of what a page is. A term in the body may be incidental; the same term in the title usually means the page is about it.

Uses:

Purpose	Pattern
Finding documents by their formal title	An exact phrase in the title

Purpose	Pattern
Finding page types	Words characteristic of that page type in titles
Filtering out passing mentions	Moving a term from the body to the title requirement
Site structure analysis	Combined with a site restriction, to see how a site names its pages

Caution. `allintitle:` is the operator that most commonly fails when combined with others. Use `intitle:` repeatedly instead.

1.3.4 Body Text Operators

Operator	Form	Meaning
<code>intext:</code>	<code>intext:"registration number"</code>	The phrase appears in the page body
<code>allintext:</code>	<code>allintext:registration number 2024</code>	All terms in the body

Uses: forcing a term to appear in the body rather than in navigation, boilerplate, or the title, which is useful when a site's template repeats a word on every page.

Caution. The effect of `intext:` is often subtle, because ordinary terms are already matched against the body. It earns its place when you need to require a term in the body while another operator constrains the title or URL.

1.3.5 Combining Structural Operators

The structural operators become powerful in combination, and the combination pattern is worth memorizing:

Goal	Pattern
An organization's own reports	Site restriction plus file type plus a title phrase
A section of a large site	Site restriction plus a URL path term
Documents of a type across a sector	A domain suffix restriction plus file type plus a title phrase
Pages a site publishes about a topic	Site restriction plus an exact phrase
Third-party coverage of an organization	Exact organization name plus exclusion of their own domain
Directory-style listings	Title and URL patterns characteristic of automatically generated index pages

On that last row, Rule 11 applies directly. Queries that surface automatically generated directory listings frequently return material that was exposed unintentionally. Understanding the query is legitimate and useful for auditing your own organization; using it against systems you do not own is not part of this course.

1.3.6 Anchor Text Operators

Operator	Form	Status
<code>inanchor:</code>	<code>inanchor:"annual report"</code>	Historically matched the text of links pointing to a page; behaviour has become unreliable
<code>allinanchor:</code>		Same, less reliable

Position. These were once useful for finding pages that others describe in a particular way. Their current behaviour is inconsistent enough that they should be tested before use and not relied upon. They are covered here because you will encounter them in older guides.

1.4 Content, File, and Source Filters

1.4.1 File Type Restriction

Operator	Form	Meaning
<code>filetype:</code>	<code>filetype:pdf</code>	Only that file format
<code>ext:</code>	<code>ext:pdf</code>	Historically equivalent; test both

This is the operator that finds documents rather than pages, and documents are where investigative value concentrates. Chapter 3 established that filings, reports, and presentations carry the substance; this is how you reach them.

Formats worth knowing:

Format	Typical content
PDF	Reports, filings, manuals, presentations, minutes, brochures
Word processing formats	Drafts, templates, letters, internal-style documents
Spreadsheet formats	Data, schedules, budgets, registers
Presentation formats	Conference talks, internal briefings, sales decks
Plain text and delimited formats	Data exports, logs, configuration
Structured data formats	Datasets, feeds, exports

The document-discovery pattern, which is the single most productive query shape in this chapter:

A site restriction or a domain suffix, plus a file type restriction, plus an exact phrase from the document's expected title or content.

Cautions.

- Google indexes the text of many document formats but not all, and coverage varies by format and by size.
- A file type restriction returns what is indexed, which is a subset of what exists on the site.
- Spreadsheet and text formats are exactly where accidental exposures concentrate, and Rule 11 applies. Chapter 8's metadata material also applies: a downloaded document carries metadata about its authors and organization, which is a defensive consideration for your own organization's published files.

1.4.2 Source Restriction for News

Operator	Form	Meaning
source:	source:"Publication Name"	Restricts news results to a publication; works within the news interface

Uses: comparing how different outlets covered the same event, and finding a specific publication's coverage when its own site search is poor. Behaviour is limited to the news corpus and it is inconsistent in the general web interface.

1.4.3 Location Restriction for News

Operator	Form	Meaning
location: or loc:	location:"city name"	Restricts news results to a location; news interface

Uses: local coverage of an event or an organization, which Chapter 3 identified as a category that global searching systematically underweights.

1.4.4 Related Sites

Operator	Form	Status
related:	related:example.com	Returned sites Google considered similar; support has been withdrawn or degraded

Position. When it worked, it was genuinely useful for mapping an industry or finding an organization's

peers. Test it; if it no longer functions, the alternatives are industry directories, trade association member lists, and the structured sources of Chapter 3.

1.4.5 Definition and Reference Commands

Operator	Form	Purpose
<code>define:</code>	<code>define:term</code>	Definitions, sometimes with usage and etymology
Unit conversion	100 miles in km	Conversion, using the word "in"
Currency conversion	Currency amounts with a target currency	Conversion at a current rate
Calculations	Arithmetic expressions	Direct evaluation
<code>weather:</code>	<code>weather:location</code>	Current conditions
<code>stocks:</code>	<code>stocks:ticker</code>	Market data
<code>map:</code>	<code>map:location</code>	Map results
<code>movie:</code>	<code>movie:title</code>	Film information

Investigative relevance is modest but real. `define:` is useful for domain jargon; conversion and calculation matter when reconciling figures across sources with different units, which is a Chapter 4 processing task; and `weather:` has an investigative use in Chapter 2's chronolocation, though authoritative historical weather records come from meteorological services rather than from a search box.

1.5 Advanced Positioning and Temporal Search

1.5.1 Proximity: AROUND

Operator	Form	Meaning
<code>AROUND(n)</code>	<code>term1 AROUND(5) term2</code>	The two terms within approximately n words of each other

Capitalization is required and the number is in parentheses. Lowercase fails.

Why proximity matters. Implicit AND requires both terms somewhere on the page, which on a long page means almost nothing. Proximity requires them in the same thought.

Situation	Why proximity helps
A name and a role, on a page listing hundreds of names	The two must be adjacent to be about the same person

Situation	Why proximity helps
An entity and an event, on a news archive page	Both terms appear on any archive page; proximity requires them in the same story
A product and a defect, in a long forum thread	Both appear; proximity requires them in one message
Two entities, testing for a relationship	Their co-occurrence in one sentence is meaningful; on one page is not

Cautions. AROUND is undocumented, its behaviour is approximate, and the word distance is not exact. It also interacts unpredictably with quotation marks. Test it before relying on it, and treat it as a strong hint rather than a precise filter.

1.5.2 Date Restriction: before and after

Operator	Form	Meaning
before:	before:2020-01-01	Content Google associates with a date before that
after:	after:2018-06-30	Content dated after that
Combined	after:2018-01-01 before:2019-01-01	A window

Format. A full ISO-style date is most reliable; year-only forms work in many cases. Test the behaviour you need.

Why this is one of the most valuable operators for investigative work. Chapter 9 established freshness bias: recent content outranks original historical material, which actively obstructs provenance tracing. Date restriction is the counter, and it is the operator that makes Chapter 5's timeline work practical.

Use	Pattern
Finding original reporting rather than aggregation	An exact phrase plus a window around the event
Establishing when a claim first appeared	Progressively earlier windows until results stop
Excluding a recent event that floods results	A before restriction
Historical research	A window many years back
Testing whether a page existed at a time	Combined with archives, per Chapter 3

The critical caution. The date Google associates with a page is inferred and is frequently wrong. It may be a publication date, a modification date, a date extracted from the page text, or a crawl date.

Date restriction is a strong filter and a weak fact. Never assert a publication date on the strength of a date operator alone; verify from the page itself, from an archive, or from a structured source.

1.5.3 The Withdrawn Date Range Operator

An older operator accepted a range expressed in a specialist date format. It was always fragile and it is no longer reliable. The **before** and **after** operators, plus the interface date-range tools, replaced it. It is mentioned here because it appears in older guides and because encountering it should tell you the guide is out of date.

1.5.4 Interface Date Tools

The search interface offers date-range filtering through its tools menu, which is sometimes more reliable than the operators and produces a filter visible in the interface. Two practical notes: the interface filter and the operators can produce different results, and the filter state is easy to forget, which produces confusing results later. **Record which you used in the search log.**

1.5.5 Combining Temporal and Structural Operators

The highest-value combination in the chapter, after document discovery:

Goal	Pattern
An organization's documents from a period	Site restriction plus file type plus a date window
The earliest appearance of a claim	Exact phrase plus progressively earlier before-restrictions
Coverage of an event, excluding later commentary	Exact phrase plus a tight date window around the event
A site's content from before a redesign	Site restriction plus a before-restriction, then archives for what is gone

1.6 Specialized Google Commands and Corpora

1.6.1 The Advanced Search Interface

Google's advanced search form exposes many operators through a graphical interface, and it is worth knowing for three reasons: it shows you the operator syntax it generates, which is a learning tool; it exposes options that are awkward to type; and it includes filters such as usage rights and region that are less commonly used from the query box.

The usage rights filter is directly relevant to Chapter 7's copyright material: it allows filtering to material with licence terms permitting reuse, which is a legitimate first step before relying on any image or document.

1.6.2 Scholarly Search

A separate corpus with its own coverage and its own operators.

Capability	Note
Author restriction	Finds work by a named author; frequently ambiguous for common names
Publication restriction	Restricts to a journal or venue
Date range	Publication years
Citation tracing	Forward citation searching, which is the single most useful feature
Related articles	Useful for expanding a literature base

Investigative relevance. Chapter 3 established that academic literature supplies method, measurement, and defensible general claims. The scholarly corpus has different coverage from the general index, so a null result in one says nothing about the other.

1.6.3 Patents

A separate corpus covering patent documents, with searching by assignee, inventor, classification, and date.

Investigative relevance. Patents disclose technical capability, organizational relationships through assignment, and the timing of development. For corporate and competitive research they are a primary source that general web search reaches poorly.

1.6.4 Books

A large digitized corpus with full-text search and limited display.

Investigative relevance. Chapter 3's point applies: full-text search across millions of volumes tells you whether a book contains what you need and where, even when you cannot read the page. For pre-digital research this is frequently the only path.

1.6.5 News

A separate corpus with the source and location operators from Section 1.4, and its own date handling.

Investigative relevance. Useful for comparative coverage analysis, with the caution that its archive depth is shallower than dedicated newspaper archives, which Chapter 3 identified as a distinct and richer category.

1.6.6 Images

Covered in Chapter 9's comparison. Two operator-adjacent points: the image interface supports filtering by size, type, colour, and usage rights, and reverse image search is a separate function rather than an operator. Chapter 2's verification sequence governs its use.

1.6.7 Programmable Search

Google offers a mechanism for building a custom search interface restricted to a defined set of sites.

For an investigator working a recurring beat, this is a legitimate way to search a curated set of sources repeatedly without retyping site restrictions.

Rule 5 note. This is a configuration mechanism, not an automation mechanism, and it does not license programmatic querying.

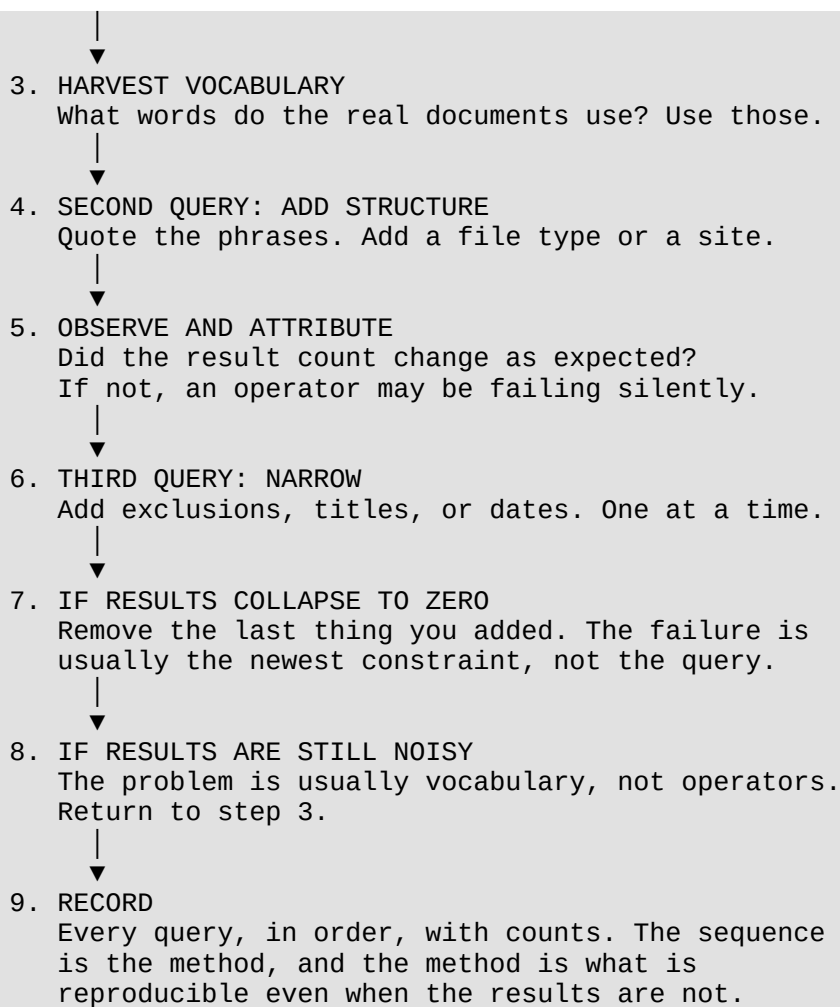
1.7 Practical Applications

1.7.1 Query Patterns Worth Memorizing

Goal	Pattern	Why it works
Find an organization's documents	Site restriction + file type + title phrase	Targets the artifact, on their own domain
Find a document type across a sector	Domain suffix + file type + title phrase	Finds institutions you did not know existed
Find third-party coverage	Exact entity name + exclusion of their domain	Removes their own material
Trace a claim's origin	Exact phrase + progressively earlier date restrictions	Defeats freshness bias
Detect copied text	An exact distinctive sentence, unrestricted	Reveals every page carrying it
Map a site's structure	Site restriction + URL path terms	Shows how the site is organized
Find an entity under name variants	Grouped OR of every variant + a context term	One query covers the variant set
Establish a relationship	Two entity names + proximity	Requires co-occurrence in a thought
Find datasets	Domain suffix + data file types + a subject phrase	Data lives in formats web search rarely surfaces
Find a person's professional role	Exact name + exact role phrase, both in quotes	Only for individuals in an official capacity, per Rule 1
Audit your own organization	Site restriction + file types + sensitive term patterns	The defensive use, per Rule 11

1.7.2 The Query Refinement Workflow

1. WRITE THE QUESTION
What page or document would answer it?
↓
2. FIRST QUERY: BROAD
Two or three content terms, no operators.
Purpose: see the shape of the material and learn the vocabulary the sources use.



Step 8 is the one investigators resist. When a query is noisy, the instinct is to add another operator. Usually the fix is a better term, taken from a document that already exists.

1.7.3 Combining Operators: What Works and What Conflicts

Combination	Works?	Note
Site restriction + file type	Yes, reliably	The workhorse
Site restriction + exact phrase	Yes	
Site restriction + date restriction	Yes	
File type + exact phrase	Yes	
Multiple <code>intitle:</code> terms	Yes	Preferred over <code>allintitle:</code>
Multiple <code>inurl:</code> terms	Yes	Preferred over <code>allinurl:</code>
<code>allintitle:</code> + other operators	Frequently fails	Use repeated <code>intitle:</code>
<code>allinurl:</code> + other operators	Frequently fails	Use repeated <code>inurl:</code>
<code>allintext:</code> + other operators	Frequently fails	Use <code>intext:</code>

Combination	Works?	Note
Exclusion + any operator	Yes	Exclude domains, file types, titles
OR + grouping + other operators	Yes, with parentheses	Group explicitly
Proximity + quotation marks	Unpredictable	Test
Multiple site restrictions	No	Use OR with grouping instead
Multiple file types	No	Use OR with grouping, or separate queries

The general rule: the `all`-prefixed operators want to own the whole query. Use the `in`-prefixed forms repeatedly instead, and you can combine freely.

1.7.4 Result Count Interpretation

The reported number of results is an estimate, and it is unreliable in specific ways:

Property	Reality
Accuracy	An approximation, sometimes off by orders of magnitude
Stability	Changes between refreshes of the same query
Deep paging	You cannot reach most of the claimed results
Comparative use	Rough comparison between two queries is the only defensible use
Reporting	Never cite a result count as a finding

What it is good for: noticing that a query went from 40,000 to 4 when you added an operator, which tells you the operator did something. That is a diagnostic, not a measurement.

1.7.5 When Operators Are the Wrong Tool

Carried from Chapter 3 and Chapter 9, and worth repeating because operator skill makes it tempting to use them for everything:

Question	Better tool
Who owns this property?	The recorder's or assessor's system
Has this entity been sued?	The court's case system
Is this entity registered?	The corporate registry
What permits exist at this address?	The permit portal
What did this page say last year?	An archive
What is the peer-reviewed evidence?	A scholarly database

Question	Better tool
What does this filing say?	The regulator's filing system

A well-constructed query against the general index is a poor substitute for a query against the system that holds the record. The operator's job is frequently to find out which system that is.

1.8 Deprecated Operators and Limitations

1.8.1 Why Operators Get Withdrawn

Operators are withdrawn because they are little used, because they are expensive to support, because they are abused, or because the underlying feature no longer exists. Google rarely announces withdrawals in advance and sometimes never announces them at all: an operator simply starts returning nothing, or starts being ignored.

The consequence for you: an operator you learned from a guide may be dead, and the guide will not tell you. Section 1.1.5's test procedure is the only reliable check.

1.8.2 Operators That Have Been Withdrawn or Degraded

As understood at the time of writing. Verify each one before relying on it, and expect this list to grow.

Operator	What it did	Status	What to use instead
The plus prefix	Forced an exact term, disabling synonyms and stemming	Withdrawn; the syntax was reused for a social product and then removed	Quotation marks around the single term
The tilde prefix	Included synonyms of a term	Withdrawn	Explicit OR with the synonyms you want
link:	Listed pages linking to a URL	Withdrawn	Backlink data from dedicated services; site-restricted searches for mentions
info:	Returned information Google held about a URL	Withdrawn	The site restriction plus a direct look at the page
cache:	Retrieved Google's stored copy of a page	Withdrawn	Web archives (Chapter 3), which are more complete and dated
related:	Found similar sites	Withdrawn or heavily degraded	Industry directories, association member lists, structured sources
daterange:	Range in a specialist	Unreliable; effectively	before: and

Operator	What it did	Status	What to use instead
	date format	obsolete	after :, or interface date tools
inanchor: and allinanchor:	Matched link text pointing to a page	Degraded and inconsistent	Test before use; treat as unreliable
phonebook: and residential lookup operators	Directory lookups of individuals	Withdrawn	Nothing, and Rule 1 means you should not want it
blogurl:	Restricted to a blog URL within the blog corpus	Withdrawn with the corpus	Site restriction
Numeric-range wildcard variants	Older range syntaxes	Superseded	The two-dot range form

The withdrawal of the cache operator deserves particular note, because it removed a capability investigators relied on daily. Its replacement is not another operator; it is the archive category from Chapter 3, and archives are better for the purpose because they are dated, independent of the engine, and preserve multiple states.

1.8.3 Structural Limitations

Beyond individual operators, five limitations constrain everything in this chapter:

Limitation	Consequence
Index dependence	No operator retrieves an uncrawled page. Rule 10
Silent failure	An unsupported or ignored operator produces plausible results that are not filtered
Result depth restriction	Most of a large result set is unreachable, so an operator that ranks a page at 400 has effectively hidden it
Approximation	Result counts, date associations, and proximity distances are all approximate
Personalization and localization	Your results are not the neutral results, and there is no way to see the difference (Chapter 9)

1.8.4 Rule 11 in Practice

The operators in this chapter are the tools used to find accidental exposures, and collections of such queries are published and maintained openly.

The legitimate uses:

Use	Why it is legitimate
Auditing your own organization's indexed footprint	You own it; Chapter 8's organizational footprint audit
Auditing a client's footprint, with written authorization	Authorized
Understanding the technique to defend against it	Defensive knowledge
Research and teaching, at the level of understanding	This chapter

The illegitimate uses:

Use	Why it is not permitted
Searching for exposed systems you do not own	Chapter 7: exposure by error is not publication
Accessing anything a query surfaces on a third party's system	Unauthorized access, regardless of how it was found
Collecting exposed personal data	Chapter 7's categorical position
Testing whether credentials found in an exposure work	Unauthorized access

If you encounter an exposure while doing something else, which happens, Chapter 7 Section 1.8 governs: stop, do not access further, record what you saw and what you did not do, and report through a coordinated disclosure process.

Part 2: Comparison Tables

A standing caveat. Operator behaviour changes without notice, and several operators are undocumented. Verify anything you intend to rely on using Section 1.1.5, and record the date of your verification.

2.1 Basic Searches vs. Advanced Searches

Dimension	Basic search	Advanced search
Query form	Two to five natural-language terms	Terms plus operators, deliberately structured
What it asks for	Pages about a topic	A specific artifact with specific properties
Typical result count	Thousands to millions	Tens to hundreds
Precision	Low	High
Recall	High	Deliberately reduced

Dimension	Basic search	Advanced search
Noise	Aggregators, commercial pages, tangential coverage	Minimal, if the operators worked
Reproducibility	Low	Higher, because the query is explicit
Diagnosis when it fails	Difficult; you cannot tell what went wrong	Straightforward; remove operators one at a time
Documentation value	Little	The query is the method
Time to first useful result	Fast to something, slow to the right thing	Slower to start, much faster to finish
Best for	Orientation; learning the vocabulary of a field	Retrieval, once you know what you want
Failure mode	Drowning in results	Over-constraining and returning nothing
Skill required	None	Learned, and this chapter is the learning

2.2 Every Operator Covered in This Chapter

Operator	Syntax	Purpose	Status	Combines well	Chief caution
Implicit AND	a b c	All terms required	Current	Everything	Common words may be dropped
Exact phrase	"phrase"	Terms in order, adjacent	Current	Everything	Typographic quotes break it
Exclusion	-term	Removes results with the term	Current	Everything, including other operators	Over-exclusion is silent
OR	a OR b or a \ b	Either term	Current	With parentheses	Lowercase or fails silently
Grouping	(a OR b) c	Controls Boolean scope	Current	With OR	Ambiguity without it
Wildcard	"a * b"	Placeholder for words	Current	Inside quotes	Loose; not partial-word
Numeric range	2015..2020	Any number in range	Current	Everything	Matches text, not data
Site restriction	site:domain	Limits to a domain, subdomain, path, or suffix	Current	Everything	Counts are approximate
URL term	inurl:term	Term in the	Current	Everything,	Rule 11

Operator	Syntax	Purpose	Status	Combines well	Chief caution
		URL		repeatable	territory
All URL terms	<code>allinurl:terms</code>	All terms in the URL	Current	Poorly	Use repeated <code>inurl:</code>
Title term	<code>intitle:term</code>	Term in the page title	Current	Everything, repeatable	The best precision operator
All title terms	<code>allintitle:terms</code>	All terms in the title	Current	Poorly	Use repeated <code>intitle:</code>
Body term	<code>intext:term</code>	Term in the page body	Current	Everything	Effect is subtle
All body terms	<code>allintext:terms</code>	All terms in the body	Current	Poorly	Use <code>intext:</code>
Anchor text	<code>inanchor:term</code>	Term in links pointing to a page	Degraded	Unreliably	Test before use
File type	<code>filetype:ext</code>	Only that format	Current	Everything	One type per query; use OR for several
Extension	<code>ext:ext</code>	Historically equivalent	Current, test	Everything	Verify equivalence
News source	<code>source:"Name"</code>	Restricts to a publication	News corpus	Within news	Not general web
News location	<code>location:place</code>	Restricts to a place	News corpus	Within news	Not general web
Proximity	<code>a AROUND(n) b</code>	Terms within about n words	Undocumented	Unpredictably with quotes	Capitalize; approximate
Before date	<code>before:YYYY-MM-DD</code>	Content dated earlier	Current	Everything	The date is inferred, often wrong
After date	<code>after:YYYY-MM-DD</code>	Content dated later	Current	Everything	Same
Definition	<code>define:term</code>	Definitions	Current	Alone	Not investigative on its own
Weather	<code>weather:place</code>	Current conditions	Current	Alone	Not an authoritative record
Stocks	<code>stocks:ticker</code>	Market data	Current	Alone	Point-in-time

Operator	Syntax	Purpose	Status	Combines well	Chief caution
Map	map:place	Map results	Current	Alone	
Movie	movie:title	Film information	Current	Alone	
Related sites	related:domain	Similar sites	Withdrawn or degraded		Use directories instead
Cache	cache:url	Google's stored copy	Withdrawn		Use web archives
Link	link:url	Pages linking to a URL	Withdrawn		Use backlink services
Info	info:url	Information about a URL	Withdrawn		Visit the page
Date range	daterange:...	Range in a specialist format	Obsolete		Use before and after
Plus prefix	+term	Forced exact term	Withdrawn		Quote the single term
Tilde prefix	~term	Synonyms	Withdrawn		Explicit OR
Phonebook	phonebook:	Directory lookup	Withdrawn		Not permitted under Rule 1 anyway

2.3 Boolean Operators and Their Uses

Operator	Syntax	Logic	Investigative use	Failure mode
Implicit AND	Space between terms	All required	Narrowing by adding a term; the default control	Common words dropped from unquoted queries
OR	OR or \ , capitalized	Either required	Covering name variants, synonyms, and alternative terminology in one query	Lowercase or treated as a word, silently
NOT (exclusion)	- term, no space	Term must be absent	Removing a dominant unrelated meaning; removing a subject's own domain	Removing relevant material invisibly
Grouping	()	Controls scope	Making a multi-term OR combine	Without it, precedence is

Operator	Syntax	Logic	Investigative use	Failure mode
			correctly with other terms	ambiguous
Exact phrase	" "	Adjacency and order	Names, titles, error strings, boilerplate, quoted claims	Typographic quotation marks
Wildcard	*	Any word in a slot	Phrase variants; discovering the word that fills a role	Loose behaviour; unreliable outside quotes
Range	. .	Numeric span	Years, quantities, prices	Matches page text, not structured values

The composition rule. Build Boolean structures in pieces and test each piece. A five-operator query that returns nothing is impossible to diagnose in one step, and the cause is nearly always the most recently added constraint.

2.4 Page Structure Operators

Operator	Targets	Precision	Best use	Combines	Caution
<code>site:</code>	Domain, subdomain, path, or suffix	High	Enumerating an organization's material; sector sweeps; exclusion	Everything	One domain per operator; use OR grouping for several
<code>inurl:</code>	The URL string	Medium to high	Finding site sections, repositories, and functional pages	Everything, repeatable	Rule 11 applies to functional URL patterns
<code>allinurl:</code>	The URL string, all terms	High	Rarely, alone	Poorly	Prefer repeated <code>inurl:</code>
<code>intitle:</code>	The page title	Highest of the structural set	Finding documents by title; filtering passing mentions	Everything, repeatable	The single best precision operator
<code>allintitle:</code>	The page title, all terms	High	Rarely, alone	Poorly	Prefer repeated <code>intitle:</code>
<code>intext:</code>	The page body	Low to medium	Forcing a term into the body while other	Everything	Effect is subtle; ordinary terms already match

Operator	Targets	Precision	Best use	Combines	Caution
			operators constrain title or URL		the body
<code>allintext:</code>	The page body, all terms	Medium	Rarely	Poorly	Prefer <code>intext:</code>
<code>inanchor:</code>	Text of inbound links	Unreliable	Historically, finding how others describe a page	Unreliably	Test; treat as degraded

The pattern to remember: title restriction tells you the page is *about* the term; body matching tells you the term merely *appears*. Moving a term from the body requirement to the title requirement is the fastest single precision improvement available.

2.5 File Search Operators

Operator	Syntax	Finds	Typical investigative content	Caution
<code>filetype:</code>	<code>filetype:pdf</code>	Documents of that format	Reports, filings, manuals, minutes, presentations	One format per query
<code>ext:</code>	<code>ext:pdf</code>	Historically equivalent	Same	Verify equivalence before relying
Combined with <code>site:</code>	<code>site:x</code> <code>filetype:y</code>	An organization's documents	The single most productive pattern in the chapter	Returns what is indexed, not what exists
Combined with a domain suffix	<code>site:.gov</code> <code>filetype:pdf</code>	Sector-wide documents	Government publications; institutional reports	Very large result sets; add a phrase
Combined with <code>intitle:</code>	<code>filetype:pdf</code> <code>intitle:"annual report"</code>	Documents by title	Named document types	
Combined with a date window	<code>filetype:pdf</code> <code>after:2020-01-01</code>	Recent documents	Time-bounded document discovery	Dates are inferred
Multiple formats	<code>(filetype:xls</code> OR <code>filetype:csv)</code>	Several formats	Data files	Grouping required

Formats and what they carry. Reports and filings in portable document format; drafts and

correspondence in word processing formats; data in spreadsheet and delimited formats; briefings in presentation formats; exports and configuration in text and structured data formats.

Rule 11 note. Spreadsheet, delimited, configuration, and backup formats are disproportionately represented in accidental exposures. Understanding the pattern is defensive knowledge; using it against systems you do not own is not permitted.

2.6 Date and Proximity Operators

Operator	Syntax	What it controls	Reliability	Best use	Caution
before:	before:2020-01-01	Upper date bound	Good as a filter	Excluding recent floods; finding original reporting	The inferred date is frequently wrong
after:	after:2018-01-01	Lower date bound	Good as a filter	Recent material; bounding a window	Same
Both combined	after:X before:Y	A window	Good	Event windows; provenance tracing	Same
Interface date tools	Tools menu	A range, or a preset period	Good, sometimes differs from operators	When operators behave oddly	Filter state persists and is easy to forget
daterange:	Specialist format	A range	Obsolete	None	Indicates an out-of-date guide
AROUND(n)	a AROUND(5) b	Word distance between terms	Approximate, undocumented	Requiring co-occurrence in a thought rather than on a page	Capitalize; unpredictable with quotes

The date caution, stated once more because it matters. A date operator filters on a date Google inferred, which may be a publication date, a modification date, a date found in the text, or a crawl date. It is a strong filter and a weak fact. Never report a publication date on the strength of a date operator; verify it from the page, an archive, or a structured source.

2.7 Current vs. Deprecated Operators

Operator	Status	If deprecated, what replaced it	How you would notice
Exact phrase, exclusion, OR, grouping, wildcard, range	Current		
site:, inurl:, intitle:, intext: and their all forms	Current		
filetype:, ext:	Current		
before:, after:	Current		
AROUND(n)	Current but undocumented		Sudden behaviour change without notice
define:, weather:, stocks:, map:, movie:	Current		
source:, location:	Current within the news corpus		Fails in the general web interface
inanchor:, allinanchor:	Degraded	Nothing equivalent	Inconsistent results across similar queries
related:	Withdrawn or heavily degraded	Directories, association lists	Empty or irrelevant results
cache:	Withdrawn	Web archives (Chapter 3)	Error or ordinary search results
link:	Withdrawn	Backlink data services	Ordinary search results, not backlinks
info:	Withdrawn	Visiting the page; the site restriction	Ordinary search results
daterange:	Obsolete	before: and after:	Unpredictable or ignored
+term	Withdrawn	Quote the term	The plus is ignored or treated as text
~term	Withdrawn	Explicit OR with chosen synonyms	The tilde is ignored
phonebook: and similar	Withdrawn	Nothing; Rule 1 prohibits the use case	Ordinary search results

How to keep this table current for yourself: test the operators you depend on quarterly using Section 1.1.5, and record the test date. A guide that lists `cache:` as available tells you the guide predates its

withdrawal, and you should be suspicious of everything else in it.

2.8 Common Syntax Errors and Corrections

Error	Example of the mistake	What happens	Correction
Space after the operator colon	site: example.com	The operator is ignored; the domain becomes a search term	No space: site:example.com
Space after the minus sign	report - draft	The minus is treated as text	No space: report -draft
Lowercase or	apple or orange	Treated as an ordinary word; results are an implicit AND	Capitalize: OR, or use \
Typing AND	apple AND orange	Redundant; may be treated as a term	Omit it
Typographic quotation marks	Curly quotes pasted from a document	The exactPhrase operator may not register	Retype straight quotes
Protocol in a site restriction	site:https://example.com	Frequently fails	Domain only: site:example.com
A full URL in a site restriction	site:example.com/page.html	Unreliable	Use a path prefix, or a inurl: term
Multiple site restrictions	site:a.com site:b.com	Returns nothing; the conditions conflict	(site:a.com OR site:b.com)
Multiple file types	filetype:pdf filetype:docx	Returns nothing	(filetype:pdf OR filetype:docx)
allintitle: with other operators	allintitle:report site:x.com	The other operator is frequently ignored	intitle:report site:x.com
Dot in a file type	filetype:.pdf	May fail	No dot: filetype:pdf
Wildcard for partial words	manufact*	Does not work as a stem wildcard	Use OR with the variants you want
Unquoted phrase with common words	the state of the report	Common words may be dropped	Quote the phrase
Ungrouped OR with other terms	apple OR orange juice	Ambiguous scope	(apple OR orange) juice
Date without a full format	before:2020	May work, may be interpreted loosely	Use before:2020-01-01 and test
Lowercase around(5)	a around(5) b	Treated as text	AROUND(5), capitalized

Error	Example of the mistake	What happens	Correction
Operators inside quotation marks	"site:example.com report"	Searched as a literal string	Keep operators outside the quotes
Trailing punctuation from a copied query	site:example.com ,	The comma becomes part of the domain	Clean the query before running it

The diagnostic habit: when a query behaves oddly, remove operators one at a time from the most recently added, and watch the result count. The operator that changes nothing when removed was not working.

2.9 Best Practices for Combining Operators

Practice	Reason
Build up, do not debug down	Add operators one at a time and observe. A five-operator query that fails cannot be diagnosed in one step
Prefer in-prefixed operators over all-prefixed ones	The all forms want the whole query and combine poorly
Use grouping whenever OR appears with anything else	Precedence is ambiguous otherwise
Put the most restrictive operator in first	Not because order changes results, but because it makes your intent legible when you re-read the query
Quote every multi-word phrase	Prevents word dropping and enforces adjacency
Test each operator's effect via the result count	The count is unreliable as a measurement and useful as a diagnostic
Keep a known-good query for each operator	For testing whether the operator still works
Log the query exactly, and screenshot the results	Chapter 6; searches are not reproducible
When results collapse to zero, remove the newest constraint	The failure is almost always the last addition
When results are noisy, improve the vocabulary before adding operators	Noise is usually a term problem, not a structure problem
Vary one thing per query	So you can attribute the change
Stop when a structured source would answer it	Chapter 3; the operator's job is often to find the system, not the record

Part 3: Case Studies

Each case explains the methodology, the operators used, why each was selected, and how it improved accuracy or efficiency. Cases marked *composite* are built from documented professional practice and are not real organizations. Query patterns are given in structural form rather than as literal strings, because literal strings age badly.

3.1 Academic Research: Building a Grey Literature Base

(*composite*)

The situation. A researcher needs the technical literature on a narrow topic in coastal engineering, including the grey literature: technical reports, standards drafts, theses, and conference material that scholarly databases index poorly.

Methodology and operators.

Step	Query structure	Why
1	Broad natural-language query, no operators	Vocabulary harvesting: learning the terms the field actually uses, which differed from the researcher's initial phrasing
2	Exact phrase of the field's preferred term, unrestricted	Establishing the shape of the material and the main publishers
3	Academic domain suffix + file type + the exact phrase	Institutional repositories, which hold theses and technical reports
4	Government domain suffix + file type + the exact phrase	Agency technical reports and standards material
5	File type restricted to presentation formats + the phrase	Conference talks, which frequently predate published papers by two years
6	Title restriction on the phrase + file type	Documents whose subject is the topic rather than mentioning it
7	Date window covering the decade before the current literature	Finding the foundational work that current papers cite without linking

How the operators improved the search. Step 1 changed the query vocabulary entirely, which mattered more than any operator. Step 3 produced eleven theses, none of which appeared in the scholarly corpus. Step 5 produced conference material two years ahead of the published literature, which is where the current state of a technical field actually lives. Step 6 removed several hundred results that merely mentioned the term.

Accuracy and efficiency. The unstructured query returned tens of thousands of results, of which the first page was commercial. The structured sequence returned under 200 across seven queries, of which roughly 60 were relevant, and it took under two hours.

Lessons.

Vocabulary harvesting before operator construction is the highest-leverage step and it is not an operator at all.

Domain suffix plus file type is the grey literature pattern. Institutional repositories publish in predictable formats on predictable domains.

Presentation formats are underused. Conference decks are current, specific, and frequently more candid than papers.

3.2 Public Records Research: Finding the System and the Document

(composite)

The situation. An investigator needs building permits, inspection records, and council decisions relating to a specific address, in a jurisdiction they have not worked in.

Methodology and operators.

Step	Query structure	Why
1	Address components + generic record terms, no operators	Learning what the jurisdiction calls its systems and records
2	Government domain suffix + terms characteristic of a records portal	Finding the systems, which is the actual first task
3	Site restriction on the identified council domain + URL term for the relevant section	Locating the correct part of a large government site
4	Site restriction + file type + exact phrase from a document series title	Finding council minutes and committee papers, which are published as documents
5	Site restriction + a date window	Bounding to the relevant period
6	Exact address string in quotes, unrestricted	Finding third-party mentions: local press, objections, forum discussion
7	Exact address + exclusion of the council domain	Removing the official material to see everything else

The critical finding. The council's own site search returned nothing useful for the address. A site restriction with a file type restriction returned four committee papers whose text contained the address,

because Google had indexed the document text while the site's own search indexed only page titles. **The engine's index of a site is frequently better than the site's own search**, which is a general and underappreciated point.

Accuracy and efficiency. Steps 1 and 2 took ten minutes and identified three systems. The records themselves came from those systems, in line with Chapter 9's point, but the committee papers came from the search engine because the council's document store was not searchable through its own interface.

Lessons.

Search the site with the engine when the site's own search is poor. This is one of the most practically useful applications of a site restriction.

The exclusion at step 7 is the fastest route to third-party material about any subject with a dominant official presence.

The engine found the systems; the systems held most of the records. Chapter 3's position, demonstrated again.

3.3 Competitive Intelligence: Mapping a Competitor's Published Footprint (*composite*)

The situation. A firm needs to understand what a competitor publishes, what technology it uses, where it operates, and what it has said publicly about its plans.

Methodology and operators.

Step	Query structure	Why
1	Site restriction alone, with a broad term	An approximate inventory of what the competitor publishes
2	Site restriction + file type, iterated across document formats	Their documents: brochures, specifications, presentations, price lists
3	Site restriction + URL terms for careers and jobs sections	Job postings, which Chapter 3 identified as the richest single source about an organization
4	Site restriction + title terms for locations, offices, and facilities	Their own statement of where they operate
5	Exact company name + exclusion of their domain	Third-party coverage, trade press, and directory listings
6	Exact company name AROUND a market or region term	Requiring the two in the same thought rather than on the same page

Step	Query structure	Why
7	Site restriction + a before-date restriction, plus archives	What they said previously, and what has been removed
8	Grouped OR of every name variant + a context term	Covering trading names and abbreviations in one query

How the operators improved the search. Step 2 produced a technical specification document naming three suppliers. Step 3 produced role descriptions naming internal systems. Step 6 was decisive: the company name and a region term co-occurred on hundreds of pages, and requiring proximity reduced it to eleven, of which two were substantive. Step 7 plus archives revealed that a facility listed two years earlier no longer appeared, which Chapter 3's archived-website material identified as a category of finding.

Accuracy and efficiency. An unstructured approach would have produced the company's marketing site and press coverage. The structured approach produced documents, roles, suppliers, locations, and a removal.

Lessons.

Proximity is the operator that converts co-occurrence into relationship, and it is the one most often missing from investigators' repertoires.

A site restriction plus iterated file types is a document inventory. Run it early on any organizational subject.

Everything found here is public and published by the subject, which is Rule 2 satisfied without effort. Nothing in this case required any measure that would trouble Chapter 7.

3.4 Website Audit: What an Organization Actually Publishes

(composite; this is the method used in Lab 3)

The situation. A communications team wants to know what their own organization has indexed, including material they had forgotten, duplicate content, and documents that should not be publicly indexed.

Methodology and operators.

Step	Query structure	Purpose
1	Site restriction alone	The approximate scale of what is indexed
2	Site restriction per subdomain	Which subdomains are indexed, including ones the team had forgotten
3	Site restriction + file type,	Every indexed document, by

Step	Query structure	Purpose
	iterated	format
4	Site restriction + URL terms for staging, test, development, and backup patterns	Non-production content that has been indexed by accident
5	Site restriction + title terms characteristic of directory listings	Automatically generated index pages
6	Site restriction + an exact phrase from a page that should be unique	Duplicate content across the site
7	Site restriction + terms characteristic of internal document markings	Documents that were never meant to be published
8	Site restriction + a before-date restriction	Old content still indexed

Findings in the composite case. Two forgotten subdomains; 140 indexed documents where the team believed there were about 40; one staging subdomain fully indexed; three directory listings; a policy document marked internal; and a page duplicated at four URLs, which was diluting the site's own search performance.

The boundary that makes this legitimate. Every query was against a site the team owns, and **no query attempted to reach anything not already indexed and publicly served.** The exercise identified material to remove and to de-index; it did not access anything restricted. This is Rule 11's legitimate side.

Accuracy and efficiency. Eight query patterns, about ninety minutes, and a remediation list the team acted on for a month.

Lessons.

Organizations consistently underestimate what they publish, usually by a factor of three or more.

Subdomains are the blind spot. Staging and legacy subdomains are indexed far more often than teams expect.

This is the defensive use of the operators in this chapter, and every organization should do it periodically.

3.5 Technical Documentation Discovery

(composite)

The situation. An engineer must find the authoritative documentation, version history, and known-

issue material for a piece of industrial equipment whose vendor's website is unhelpful.

Methodology and operators.

Step	Query structure	Why
1	Exact model designation in quotes	The model string is distinctive; quoting prevents it being broken up
2	Model string + file type restricted to document formats	Manuals and specification sheets, which vendors publish as documents
3	Model string + title restriction on manual-type words	Documents whose subject is the equipment
4	Model string + a domain suffix for the vendor's home market	Original-language documentation, frequently more detailed (Chapter 9)
5	Model string + terms characteristic of technical discussion, excluding retail domains	Forum and mailing list material, where real-world behaviour is discussed
6	Model string + a date window covering the product's release period	Original documentation rather than later resale listings
7	Model string + grouped OR of related model designations	Covering a product family whose numbering varies

How the operators improved the search. The unrestricted model string returned resale listings almost exclusively. The file type restriction at step 2 cut through them immediately, because resale listings are pages and manuals are documents. Step 5's exclusion of retail domains was the second largest improvement. Step 6 addressed freshness bias, which for a product released years ago buries the original material entirely.

Accuracy and efficiency. Two of the four decisive documents were on third-party sites, not the vendor's, which is common for older equipment.

Lessons.

The file type restriction is the fastest way past commercial noise, because commerce publishes pages and engineering publishes documents.

Distinctive strings are the best queries. A model designation, a part number, or an error message is worth more than any amount of descriptive language.

Exclusion of retail domains is a reusable component for any product-related search.

3.6 Investigative Journalism: Establishing When a Claim First Appeared

(composite)

The situation. A reporter must establish the origin of a statistic circulating in policy debate, and whether it is supported.

Methodology and operators.

Step	Query structure	Why
1	The exact sentence containing the statistic, in quotes, unrestricted	Provenance tracing: finding every page carrying the identical wording
2	The same phrase + progressively earlier before-date restrictions	Walking backwards until results stop, which brackets the earliest indexed appearance
3	The phrase + a government domain suffix	Whether any official source uses the same wording
4	The phrase + file type restricted to documents	Whether the claim originates in a report rather than a page
5	The numeric value + a proximity operator with a subject term	Catching restatements that do not use the identical sentence
6	The originating organization's site restriction + the phrase	Their own current use of it
7	Archives of the earliest page	What the page said before it was edited (Chapter 3)

Findings in the composite case. Fifty-one pages carried the identical sentence. Date-bracketing placed the earliest indexed appearance on an advocacy site. The document search at step 4 found a report using the same figure with a citation to a survey. Step 7 revealed that the advocacy page's earlier version had cited the survey and that the citation had since been removed. The survey, when retrieved, measured something narrower than the claim asserted.

How the operators improved the search. Step 1 made circular reporting visible in one query. Step 2 defeated freshness bias, which otherwise buries a two-year-old original under hundreds of recent restatements. Step 5 caught paraphrased restatements that step 1 missed.

Accuracy and efficiency. The whole provenance chain took under three hours, of which the archives step was the decisive one and the only part not performed with operators.

Lessons.

Exact-phrase search across an index is the cheapest circular reporting detector in existence.

Date-bracketing is a technique, not just an operator. Progressively earlier restrictions, until results stop, is a method.

The decisive item came from an archive. Chapter 3's point, again, in a chapter about operators.

3.7 Cybersecurity Research on Authorized Environments

(composite; conducted with written authorization, per Rule 11)

The situation. A security team is authorized in writing to assess their own organization's external footprint, including what is indexed about their domains. **The authorization is what makes this exercise permissible**, and it defines its scope.

Methodology and operators.

Step	Query structure	Purpose
1	Site restriction on each owned domain and subdomain, enumerated first from their own asset inventory	Establishing what is indexed for assets they know about
2	Site restriction + file type across document and data formats	Indexed documents, including any that should not be public
3	Site restriction + URL and title terms characteristic of administrative interfaces	Whether any management interface is indexed, which is a configuration finding
4	Site restriction + terms characteristic of error pages and stack traces	Whether error output containing internal detail has been indexed
5	Site restriction + terms characteristic of directory listings	Whether any directory is browsable and indexed
6	Organization name + terms characteristic of exposure discussion, excluding their own domain	Whether third parties have published anything about them
7	Site restriction + a before-date restriction	Legacy content still indexed

Findings in the composite case. One indexed administrative login page on a forgotten subdomain, three indexed documents containing internal contact details, one indexed error page revealing a software version, and one legacy subdomain from a discontinued project.

What the team did not do, and this is the point of the case: they did not attempt to log in to the administrative interface, did not access anything beyond what the search results displayed, did not test the software version for known issues against the live system, and did not run any query against any domain they do not own. **All remediation was performed on their own systems through their own**

change process.

Lessons.

Written authorization defines the exercise, and its scope is the domain list, not "anything we find interesting."

The defensive version of these operators is genuinely valuable and every organization should run it periodically.

Finding an exposure is a configuration finding, not an invitation. The team's discipline of not touching what they found is exactly what Rule 11 requires, and it applies with equal force to systems you do own, because touching them outside change control is its own problem.

3.8 Cross-Case Comparison

Case	Decisive operator	Why it was decisive	What an unstructured search would have returned
Grey literature	Domain suffix + file type	Institutional repositories publish documents on predictable domains	Commercial pages and abstracts
Public records	Site restriction + file type	The engine indexed document text the site's own search could not	Nothing useful
Competitive footprint	Proximity	Converted co-occurrence into relationship	Marketing pages and press coverage
Website audit	Site restriction per subdomain	Revealed forgotten and staging subdomains	Nothing; this cannot be done without operators
Technical documentation	File type restriction	Separated engineering documents from retail pages	Resale listings
Claim provenance	Exact phrase + date bracketing	Made circular reporting visible and defeated freshness bias	Fifty-one restatements, ranked by recency
Authorized security audit	Site restriction + URL and title patterns	Surfaced indexed material the team did not know about	Their marketing site

Two patterns run through all seven. **The site restriction and the file type restriction appear in six of the seven**, which is why they are the two operators to learn first. And **in five of seven, the largest single improvement came from vocabulary or exclusion rather than from adding a structural operator**, which is the chapter's quiet lesson: operators sharpen a good query and cannot rescue a bad

one.

Part 4: Chapter Summary

An operator filters and reorders what is already in the index. It cannot retrieve a page that was never crawled, never indexed, or has been dropped, which is Chapter 9's Rule 10 restated at the query level. Operators change precision, not existence. The measurable effect is usually an order of magnitude: a query returning forty thousand results becomes one returning thirty, with the useful items on the first page.

Precision and recall trade against each other, and every operator choice moves along that line. Start with high recall to see the shape of the material and to learn the vocabulary the sources use, then raise precision to work it. Investigators who start narrow and stay narrow miss what they did not think to ask for.

Many operators are undocumented and none is guaranteed. Test the operators you depend on using a query with a known answer, and in particular construct a query that must return nothing if the operator is working, because **silent failure is the characteristic operator error**: an ignored operator produces results that look filtered because you expected them to be.

Boolean logic in Google is implicit AND, exact phrase with straight quotation marks, exclusion with an unspaced minus, OR in capitals, and parentheses for grouping. Lowercase OR fails silently and is one of the most common errors in practice. Wildcards work most reliably inside quoted phrases. Numeric ranges match text on the page rather than structured values.

Structural operators are the ones that turn a search into a targeted retrieval. The site restriction is the most useful single operator in the chapter, for enumerating an organization's material, sweeping a sector by domain suffix, auditing, and excluding a dominant domain. Title restriction is the most precise: a term in a title usually means the page is about it, while the same term in the body may be incidental. The `all`-prefixed forms want to own the whole query and combine poorly, so use the `in`-prefixed forms repeatedly instead.

The file type restriction finds documents rather than pages, and documents are where investigative value concentrates. A site restriction or a domain suffix, plus a file type, plus an exact phrase from the expected title, is the single most productive query shape in this chapter.

Date restriction is the counter to freshness bias, which otherwise buries original reporting under recent restatement and makes provenance tracing impossible. It is a strong filter and a weak fact: the date is inferred and frequently wrong, so never assert a publication date on the strength of a date operator alone. Proximity converts co-occurrence on a page into co-occurrence in a thought, which is the difference between two terms appearing somewhere and appearing together meaningfully.

Several once-standard operators have been withdrawn, including the cache, link, info, plus-prefix, tilde-prefix, and directory-lookup operators, with the related and anchor operators degraded. The withdrawal of the cache operator matters most, and its replacement is not another operator but the

archive category from Chapter 3, which is better for the purpose because archives are dated, independent of the engine, and preserve multiple states. A guide that still lists withdrawn operators tells you it is out of date.

Five structural limitations constrain everything here: index dependence, silent failure, result depth restriction, approximation in counts and dates and distances, and personalization that you cannot see around.

The refinement workflow is broad first, harvest vocabulary, add structure, observe, narrow one constraint at a time, and back off the newest constraint when results collapse. When results are noisy the problem is usually vocabulary rather than structure, and the fix is a better term taken from a document that already exists. Build queries up rather than debugging them down, because a five-operator query that returns nothing cannot be diagnosed in one step.

Result counts are diagnostics, not measurements. They are approximate, unstable between refreshes, and mostly unreachable through paging. Their one defensible use is noticing that a query changed by an order of magnitude when you added an operator, which tells you the operator did something.

Operators are frequently the wrong tool. Ownership, litigation, registration, permits, prior page states, and peer-reviewed evidence all live in structured systems that answer definitively, and a well-constructed query against the general index is a poor substitute. The operator's job is often to find out which system holds the record and then to get out of the way.

And Rule 11 governs the whole chapter: an operator that finds something does not authorize accessing it. These operators are extremely good at surfacing material published by accident, and Chapter 7's position is unchanged: exposure by error is not publication. The legitimate uses are auditing your own footprint, auditing a client's with written authorization, and understanding the technique defensively. The case studies demonstrate the boundary, and the authorized security audit demonstrates it most directly, because what that team did not do is the substance of the case.

Part 5: Key Terms

Grouped by theme. Terms carried from earlier chapters are marked and restated briefly.

5.1 Concepts

Search operator. A special instruction in a query that constrains or modifies how the engine matches and returns results.

Documented operator. One Google publishes support for; more stable, though not guaranteed.

Undocumented operator. One in common use without published support; can change or vanish without notice.

Deprecated operator. One that has been withdrawn or degraded, whether or not the withdrawal was announced.

Silent failure. An operator being accepted and ignored, producing results that appear filtered and are not.

Precision. The proportion of returned results that are relevant.

Recall. The proportion of relevant material that was returned.

Query refinement. The iterative process of improving a query by changing one variable at a time.

Vocabulary harvesting. Running a broad query first to learn the terms the sources actually use, before adding structure.

Artifact-oriented searching (Ch.9). Searching for the document that would contain the answer rather than for the topic.

Result count. The reported estimate of matching pages; a diagnostic, not a measurement.

Result depth restriction (Ch.9). The limit on how far into a result set a user can page.

Index dependence. The principle that no operator can return a page the crawler never fetched.

5.2 Boolean and Fundamentals

Implicit AND. Google's default treatment of multiple terms as all required.

Exact phrase. Straight double quotation marks requiring terms in order and adjacent.

Straight versus typographic quotation marks. The distinction that breaks copied queries, because curly quotes may not register as the operator.

Exclusion. An unspaced minus before a term, word, phrase, or operator, removing matching results.

Over-exclusion. Removing relevant material invisibly through excessive exclusion.

OR. Capitalized, or the pipe character, requiring either term. Lowercase fails silently.

Grouping. Parentheses controlling how OR combines with implicit AND.

Wildcard. An asterisk standing for one or more words, most reliable inside a quoted phrase.

Numeric range. Two dots between numbers, matching any value in the span as text on the page.

Stop word handling. The inconsistent treatment of very common words in unquoted queries.

5.3 Structural Operators

Site restriction (site:). Limits results to a domain, subdomain, path, or domain suffix.

Domain suffix sweep. Using a site restriction on a top-level or sector suffix to find institutional material across many organizations.

URL operator (inurl:, allinurl:). Requires a term in the page URL.

Title operator (intitle:, allintitle:). Requires a term in the page title; the highest-precision structural operator.

Body operator (**intext:**, **allintext:**). Requires a term in the page body.

Anchor operator (**inanchor:**). Historically matched the text of links pointing to a page; degraded.

The all prefix problem. The tendency of **allintitle:**, **allinurl:**, and **allintext:** to combine poorly with other operators.

Directory listing. An automatically generated index page; frequently indicative of unintended exposure.

5.4 File, Source, and Specialized

File type restriction (**filetype:**, **ext:**). Limits results to a document format.

Document discovery pattern. Site or domain suffix, plus file type, plus an exact phrase; the most productive query shape in the chapter.

News source restriction (**source:**). Restricts news results to a publication.

News location restriction (**location:**). Restricts news results to a place.

Definition command (**define:**). Returns definitions.

Reference commands. Weather, stocks, map, and film commands returning structured answers.

Usage rights filter. An advanced search option filtering by licence terms; relevant to Chapter 7's copyright material.

Specialized corpus (Ch.9). A separate index for scholarly literature, patents, books, or news, with its own coverage and operators.

Programmable search. A configuration mechanism for building a search interface restricted to a defined set of sites.

5.5 Temporal and Positional

Before and after operators. Upper and lower date bounds, filtering on a date Google inferred.

Inferred date. The date Google associates with a page, which may be a publication, modification, extracted, or crawl date, and is frequently wrong.

Date bracketing. Progressively earlier date restrictions until results stop, used to find the earliest indexed appearance of a claim.

Freshness bias (Ch.9). Ranking preference for recent content, which buries original historical material.

Interface date tools. The tools-menu date filter, which can behave differently from the operators and whose state persists.

Proximity operator (**AROUND(n)**). Requires two terms within approximately n words; capitalized, undocumented, approximate.

Co-occurrence versus adjacency. The distinction between two terms appearing on the same page and

appearing in the same thought.

5.6 Discipline and Boundaries

Operator testing. Verifying that an operator works, including with a query that must return nothing if it is functioning.

Known-good query. A saved query with a verified answer, kept for testing whether an operator still works.

Build up, do not debug down. Adding operators one at a time so that failures are attributable.

Newest-constraint rule. When results collapse to zero, the failure is almost always the most recently added operator.

Search log (Ch.6, Ch.9). The record of every query with its exact string, operators, locale, state, time, count, and outcome.

Rule 11. An operator that finds something does not authorize accessing it.

Accidental exposure (Ch.7). Material published by mistake, which is not public and must not be collected.

Authorized audit. An assessment of an organization's own indexed footprint, or a client's under written authorization, using read-only public queries.

Read-only public query. A query against indexed content that makes no attempt to reach restricted resources.

Part 6: Knowledge Assessment

6.1 Short Answer (20 questions)

Answer in three to five sentences unless otherwise noted.

1. Explain mechanically what an operator does, and why no operator can expand the index.
2. Define precision and recall, and describe the working pattern for moving between them.
3. Explain the difference between a documented and an undocumented operator, and the risk each carries.
4. Describe the procedure for testing whether an operator still works, and explain why step five of that procedure is essential.
5. Explain silent operator failure and give an example of how you would detect it.
6. Explain how Google implements AND, OR, and NOT, and identify the most common syntax error among the three.
7. Explain why exact phrase search is the highest-yield operator in investigative work, with four

use cases.

8. Explain what exclusion can be applied to beyond individual words, and describe the risk of over-exclusion.
9. Explain why the site restriction is the most useful single operator, with four distinct uses.
10. Explain why a term in a page title carries more weight than the same term in the body, and what that means for query construction.
11. Explain the `all` prefix problem and state the recommended alternative.
12. Describe the document discovery pattern and explain why it is the most productive query shape in the chapter.
13. Explain why date restriction is valuable for provenance work, and state the caution that applies to inferred dates.
14. Explain the difference between co-occurrence on a page and co-occurrence in a thought, and which operator addresses it.
15. Name five operators that have been withdrawn or degraded, and state what replaced each where anything did.
16. Explain why the withdrawal of the cache operator mattered, and what fills the gap.
17. Explain what a result count is good for and what it is not.
18. Describe the query refinement workflow, and explain the rule that applies when results collapse to zero.
19. Explain when operators are the wrong tool, with four examples of questions better answered elsewhere.
20. State Rule 11 and explain the boundary between legitimate and illegitimate uses of exposure-surfacing queries.

6.2 Multiple Choice (15 questions)

1. An operator's function is to: a) Expand the set of pages the engine can return b) Filter and reorder pages already in the index c) Force the crawler to fetch new pages d) Bypass ranking entirely
2. Which query is written correctly? a) `site: example.com annual report` b) `site:example.com "annual report"` c) `site:https://example.com annual report` d) `"site:example.com annual report"`
3. Typing `apple or orange` returns: a) Pages containing either term b) Pages containing both terms, with `or` treated as an ordinary word c) An error d) Pages containing neither
4. The most precise structural operator for establishing that a page is *about* a term is: a) `intext:` b) `inurl:` c) `intitle:` d) `inanchor:`

5. `allintitle:report site:example.com` frequently fails because: a) `site:` is deprecated b) The `all` prefixed operators combine poorly with others c) The query is too short d) Titles are not indexed
6. The most productive query shape for finding an organization's documents is: a) The organization's name alone b) Site or domain suffix restriction, plus file type, plus an exact phrase c) `related:` on their domain d) A proximity query on their name
7. A date operator filters on: a) The verified publication date b) A date Google inferred, which may be publication, modification, extracted, or crawl date c) The date the page was last visited by any user d) The domain registration date
8. `AROUND(5)` requires: a) Exactly five words between the terms b) The terms within approximately five words of each other, with the operator capitalized c) Five occurrences of each term d) A quoted phrase on both sides
9. Which operator has been withdrawn? a) `filetype:` b) `intitle:` c) `cache:` d) `before:`
10. The replacement for the withdrawn `cache` operator is: a) `info:` b) `related:` c) Web archives, which are dated and independent of the engine d) The interface date tools
11. A reported result count is best used for: a) Citing the scale of coverage in a report b) Noticing that an operator changed the result set by an order of magnitude c) Determining how many pages exist on a topic d) Comparing two engines' index sizes
12. When a query that previously worked suddenly returns nothing after you add an operator, you should first: a) Change the search engine b) Remove the most recently added constraint c) Add another operator to compensate d) Clear your browser cookies
13. Two site restrictions in one query, as in `site:a.com site:b.com`, will: a) Return results from both domains b) Return nothing, because the conditions conflict; use grouped OR instead c) Return results from whichever is listed first d) Be interpreted as an exclusion
14. Under Rule 11, finding an exposed document through a search query means: a) It is public and may be collected b) It may be collected if it is not marked confidential c) It was not published, and finding it does not authorize accessing it d) It may be collected if the finder reports it afterwards
15. In the authorized security audit case study, the significant fact was: a) How many findings the team produced b) What the team did not do after finding them c) Which operators produced the most results d) The speed of the audit

6.3 True or False (10 questions)

Mark true or false. For each false statement, write one sentence explaining what makes it false.

1. A well-constructed query can retrieve a page that Google has never crawled.
2. Adding an operator to a query always increases precision and decreases recall.
3. Lowercase `or` functions identically to capitalized `OR`.

4. Typographic quotation marks pasted from a word processor may prevent the exact-phrase operator from working.
 5. `allintitle:` combines reliably with `site:` and `filetype:`.
 6. The date associated with a page by Google is a verified publication date.
 7. A result count is an accurate measurement of how many pages match a query.
 8. Every operator described in older Google search guides still works today.
 9. Exclusion can be applied to domains, file types, and titles, not only to individual words.
 10. Finding an exposed configuration file through a search query authorizes downloading it for analysis.
-

Part 7: Critical Thinking Exercises

Selecting the appropriate operators

7.1 For each objective, name the two or three operators you would use and the reason for each: (a) finding a university's published research funding disclosures; (b) determining whether a specific sentence appears anywhere other than one site; (c) finding technical manuals for equipment released twelve years ago; (d) finding local newspaper coverage of a council decision; (e) establishing what a company published about its locations before a redesign; (f) auditing which of your own organization's documents are indexed.

7.2 An investigator asks for "everything about" a mid-sized manufacturer. Convert this into three specific queries with operators, each answering a different question, and explain what each excludes.

7.3 You need a dataset rather than an article. Describe the operator combination you would use and explain why an unrestricted query would fail.

Improving poorly written queries

7.4 Repair each and explain the error: (a) `site: gov.uk flood report`; (b) `"annual report" filetype:.pdf`; (c) `apple or orange juice recall`; (d) `allintitle:safety inspection site:example.org`; (e) `report - draft 2023`; (f) `site:a.gov site:b.gov budget`.

7.5 A colleague's query returns two million results. Diagnose the likely problems without seeing the query, and describe the three changes you would try in order.

7.6 A query returns zero results and the colleague concludes the material does not exist. Write the response, covering Rule 10, the newest-constraint rule, and the diagnostic sequence.

Combining operators effectively

7.7 Construct a single query, in structural form, that finds documents of a specific format, on a sector domain suffix, whose titles contain a phrase, published within a date window, excluding one dominant domain. Then explain what could go wrong with it and how you would test each component.

7.8 You need to cover an entity that appears under four names, in documents of two formats, on any domain. Write the query structure and explain the grouping.

7.9 Explain why `filetype:pdf filetype:docx` returns nothing and what to write instead, then generalize the principle to two other operators.

Explaining differing results

7.10 The same query returns different results for two colleagues on the same day. Give four explanations and the check for each.

7.11 Adding `intitle:` to a query reduced results from 12,000 to 6. Explain what that tells you about the material, and whether it is a good or bad outcome.

7.12 A query with an operator returns almost the same result set as the same query without it. Give two explanations and describe how you would distinguish them.

Evaluating and refining search quality

7.13 Define, for a search task of your choosing, what "good enough" looks like: how many relevant results, at what precision, and when you would stop. Then explain how you would know you had reached it.

7.14 You have run twelve queries and found nothing. Write the negative finding properly, with its coverage boundary, and state the three things you would try before concluding the material is not on the indexed web.

7.15 Take any query you have written in the last week. Improve it using at least three operators, predict the effect of each on the result count, run it, and record where your prediction was wrong.

Part 8: Hands-On Lab Exercises

General Lab Guidance

Rules 1 through 11 apply. In particular: **Rule 11** governs every query in Labs 3 and 4, **Rule 5** prohibits any automated querying, and **Rule 1** means no query in any lab targets a private individual.

Search conditions for all live labs. Use a dedicated browser profile, signed out, with locale and language settings recorded (Chapter 8). Log every query to the Section 1.1.6 standard. Screenshot any results page you rely on.

What you must not do in any lab. Attempt to access anything a query surfaces on a system you do

not own; log in to anything; download anything from a third party's exposed location; or run queries against domains you have neither authorization for nor a legitimate research interest in. Lab 3 defines its boundary explicitly.

Rubric scale. 0 to 4 per criterion: 4 professional, 3 competent, 2 developing, 1 attempted, 0 missing or fundamentally incorrect.

Lab 1: Operator by Operator

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Foundational

Objectives

1. Execute each operator individually and observe its actual behaviour.
2. Document purpose, syntax, expected result, observed result, and limitations for each.
3. Detect operators that are deprecated, degraded, or failing silently.
4. Build a personal, tested operator reference.

Prerequisites

- Sections 1.1 through 1.6, and comparison tables 2.2 and 2.7.
- A dedicated browser profile, signed out.

The Method for Each Operator

For every operator on the list, follow this five-step procedure. **The procedure is what is graded, not the results.**

1. **PREDICT**
Before searching, write what you expect the operator to do and what you expect to see.
↓
2. **BASELINE**
Run a query **WITHOUT** the operator. Record the result count and one identifiable result.
↓
3. **APPLY**
Run the same query **WITH** the operator. Record the count and whether your identifiable result survived.
↓
4. **NEGATIVE TEST**
Construct a query that **MUST** return nothing if the operator is working. Run it. Record what happened.
↓

5. RECORD

Purpose, syntax, expected, observed, works or fails, limitations, and the test date.

The Operator List

Test every operator you can. Record any you cannot test and why.

Group	Operators
Boolean and fundamentals	Exact phrase, exclusion, OR, pipe form of OR, grouping, wildcard, numeric range, implicit AND behaviour with a common word
Structure	site: on a domain, site: on a subdomain, site: on a suffix, -site:, inurl:, allinurl:, intitle:, allintitle:, intext:, allintext:, inanchor:
File and source	filetype:, ext:, filetype: combined with site:, two file types combined with OR
Temporal and positional	before:, after:, both combined, AROUND(n), AROUND(n) lowercase, interface date tool
Specialized	define:, weather:, stocks:, unit conversion, arithmetic
Deprecated (test to confirm)	cache:, link:, info:, related:, +term, ~term, daterange:

The deprecated group is not busywork. Confirming for yourself that an operator no longer works, and recording what happens instead, is what makes the rest of your reference trustworthy.

Steps

Step 1 (15 min). Set and record conditions on Worksheet 1A: profile, signed-out state, locale, language, date, time zone.

Step 2 (110 min). Work through the operator list, completing Worksheet 1B for each: operator, syntax as entered, purpose, predicted behaviour, baseline count, applied count, whether the identifiable result survived, negative test and its outcome, verdict, and limitations observed.

Step 3 (20 min). Deprecated confirmation on Worksheet 1C: for each deprecated operator, what actually happened when you ran it, and what you would use instead.

Step 4 (20 min). Combination testing on Worksheet 1D: test six combinations from table 2.9, including at least two that the table predicts will fail. Record whether the prediction held.

Step 5 (15 min). Build your personal reference card: one page, the operators you verified as working, their syntax, one use case each, and the test date. This is the artifact you will actually keep.

Step 6 (10 min). Write the anomalies note: every operator that behaved differently from this chapter's description, with what you observed. **This is graded positively.** The chapter is a snapshot and you are checking it.

Worksheet 1A: Conditions

Element	Value
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Worksheet 1B: Operator Test Record

Operator	Syntax entered	Purpose	Predicted	Baseline count	Applied count	Known result survived?	Negative test and outcome	Verdict	Limitations
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Worksheet 1C: Deprecated Confirmation

Operator	What happened	Alternative
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Worksheet 1D: Combination Testing

Combination	Predicted by table 2.9	Observed	Prediction held?
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Expected Outcomes

- **Prediction recorded before testing** for every operator. Students who fill in predictions afterwards produce suspiciously accurate ones, and the discipline is the point.
- **Negative tests actually constructed.** This is the criterion that separates a competent submission from a list of syntax. A negative test for a file type restriction, for example, uses a format that is exceptionally rare on the web.
- **The all prefixed operators should be observed failing in combination,** confirming table 2.9 from direct observation rather than from the chapter's assertion.
- **Lowercase AROUND (n) and lowercase or should both be observed failing silently,** which is the most instructive single finding in the lab.
- **The deprecated operators should be confirmed as non-functional,** with what happens instead recorded. Students frequently find that a deprecated operator returns ordinary search results rather than an error, which is exactly why silent failure is dangerous.
- **At least one anomaly** relative to this chapter's description. The chapter was written at a point in time and something will have changed.

- A personal reference card that is genuinely one page and genuinely usable.

Discussion Questions

1. Which operator behaved most differently from your prediction?
2. Which negative test was hardest to construct, and why does that matter?
3. Which deprecated operator failed most quietly, and what would that have cost an investigator who trusted it?
4. Did any combination fail that table 2.9 predicted would work, or vice versa?
5. What is on your reference card, and what did you deliberately leave off?

Grading Rubric: Lab 1 (56 points)

Criterion	Weight	4 points looks like
Conditions recorded	x1	All fields
Predictions recorded before testing	x3	Every operator, written first
Baseline and applied comparison	x3	Counts and known-result survival for every operator
Negative tests constructed	x4	Genuine must-return-nothing tests, not token queries
Coverage of the operator list	x3	All groups attempted; untestable ones recorded with reasons
Deprecated confirmation	x3	Each tested; actual behaviour recorded; alternatives named
Combination testing	x3	Six combinations, including predicted failures
Silent failure observations	x3	Lowercase OR and lowercase AROUND observed failing
Personal reference card	x2	One page, usable, dated
Anomalies note	x3	At least one genuine difference from the chapter, recorded specifically

Total possible: 112 weighted. Passing: 77.

Lab 2: Combining Operators for Document Discovery

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Combine multiple operators to locate documents of several types.
2. Justify each operator choice against the material sought.
3. Compare the yield of successive refinements.
4. Produce a documented query set another researcher could reuse.

Prerequisites

- Sections 1.2 through 1.7, and comparison tables 2.5 and 2.9.
- Lab 1 complete; bring the reference card.

The Project

The project is fictional; the searching is real, because the topic is real.

Meadowbank Trust is a fictional small charity preparing a briefing on **urban flood resilience in mid-sized cities**. You are its researcher. The trust needs, within one working day:

Deliverable	What it means
Technical PDFs	Engineering and planning guidance documents
Presentations	Conference material showing current practice
Datasets	Structured data on flood incidents, drainage, or land use
Government publications	Official guidance, strategies, and evaluations
Academic research	Peer-reviewed and grey literature
A local example	Material about one specific mid-sized city, chosen by you

Choose a real topic focus and a real city. Nothing in this lab targets a person or a private organization; the subjects are public policy, public bodies, and published research.

Steps

Step 1 (20 min). Vocabulary harvesting on Worksheet 2A. Run three broad queries with no operators. Record the terms the real documents use, which will differ from your initial phrasing. Minimum eight harvested terms, each with the source that used it.

Step 2 (25 min). Query design on Worksheet 2B. Before searching, design one query per deliverable. For each: the operators chosen, the reason for each operator, and the predicted result type. Six queries minimum.

Step 3 (60 min). Execute and refine. For each deliverable, run your designed query, then refine it at least twice. Record every query on Worksheet 2C with its result count and a yield assessment (how

many of the first ten results were the kind of thing you wanted).

Step 4 (25 min). Refinement analysis on Worksheet 2D. For each deliverable: which refinement produced the largest improvement, and was it an operator change or a vocabulary change? **Count how many of your six were vocabulary rather than operators.**

Step 5 (20 min). Cross-check on Worksheet 2E. For two deliverables, run an equivalent query on a second search engine (Chapter 9) and record what it returned that Google did not, and vice versa.

Step 6 (20 min). Produce the reusable query set: one page listing your six final queries in structural form, with a one-line note on what each finds and what it excludes, written so that a colleague could adapt them to a different topic.

Worksheet 2A: Vocabulary Harvest

Term harvested	Source that used it	Replaces my term
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Worksheet 2B: Query Design

Deliverable	Designed query (structural)	Operators and reasons	Predicted result type
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Worksheet 2C: Execution Log

#	Deliverable	Exact query	Result count	Yield in first ten	Refinement made	Why
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Worksheet 2D: Refinement Analysis

Deliverable	Largest improvement	Operator or vocabulary?	Effect on count	Effect on yield
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Worksheet 2E: Cross-Engine Check

Deliverable	Second engine	Unique to Google	Unique to the other
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Expected Outcomes

- **Eight-plus harvested terms**, with at least three that materially differ from the student's initial phrasing. Vocabulary harvesting is graded heavily because it is the step students skip.
- **Six designed queries with per-operator reasons.** A query listing operators without reasons scores 2.

- **The document deliverables should use the document discovery pattern:** a domain suffix or site restriction, plus a file type, plus a title or exact phrase.
- **The dataset deliverable should use grouped OR across data file types,** which is the case where a single file type restriction fails.
- **The government deliverable should use a domain suffix,** and students should observe that this finds bodies they did not know existed.
- **In the refinement analysis, at least two of six improvements should be vocabulary rather than operators.** This is the lab's central finding and it is stated as an expected outcome because students consistently expect the opposite.
- **The cross-engine check should produce at least one item unique to each engine,** confirming Chapter 9 from direct observation.
- A reusable query set written in structural form, adaptable to another topic.

Discussion Questions

1. How many of your six improvements came from better terms rather than better operators?
2. Which deliverable was hardest to find, and was that a query problem or an index problem?
3. The dataset query required grouped OR. Explain why to someone who has not read this chapter.
4. What did the second engine return that Google did not?
5. Your query set is meant to be reusable. What would break if you changed the topic?

Grading Rubric: Lab 2 (60 points)

Criterion	Weight	4 points looks like
Vocabulary harvesting	x4	Eight-plus terms with sources; genuine differences from initial phrasing
Query design before execution	x3	Six queries designed with per-operator reasons
Document discovery pattern applied	x3	Site or suffix, plus file type, plus phrase
Dataset query with grouped OR	x2	Correctly grouped; single restriction failure understood
Execution log completeness	x3	Every query, count, yield, refinement, and reason
Refinement analysis	x4	Operator versus vocabulary distinguished; count reported
Cross-engine check	x2	Both directions, with unique items

Criterion	Weight	4 points looks like
Reusable query set	x3	Structural, annotated, genuinely adaptable

Total possible: 96 weighted. Passing: 66.

Lab 3: Website Content Audit

Time required: 2 to 2.5 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Audit a public website's indexed footprint using read-only public queries.
2. Identify indexing issues, duplicate content, and publicly accessible documents.
3. Produce a remediation-oriented findings list.
4. Observe Rule 11's boundary in practice.

Prerequisites

- Sections 1.3, 1.4, 1.7, and 1.8.4, and case study 3.4.
- Lab 1 complete.

Subject Selection and the Boundary

Choose one of these, in this order of preference:

Preference	Subject	Why
1	A website you own or administer	No boundary questions at all
2	Your institution's or employer's site, with permission recorded	The realistic professional case
3	A large public-sector or institutional website: a government department, a university, a municipality, a public agency	Public bodies publish extensively and expect their published material to be examined

Do not choose a small business, an individual's site, a site belonging to someone you know personally, or any site where the audit could reasonably be experienced as targeting.

The absolute boundary for this lab, and it is graded:

Permitted	Not permitted
Search queries against the public index	Visiting anything that requires credentials
Viewing results and publicly served pages	Attempting to access anything a result suggests exists but does not display
Recording what is indexed	Downloading material that appears to be exposed in error
Noting that something appears indexed that probably should not be	Testing whether a login works, or probing any interface
Reporting findings to the site owner	Publishing findings about a third party's site

If your audit surfaces something that looks like an accidental exposure, stop, record what you saw and what you did not do, do not access it, and follow Chapter 7's coordinated disclosure guidance. Note it in the lab as a finding about your own process.

Steps

Step 1 (15 min). Scope and permission on Worksheet 3A: the site, why it is an appropriate subject, the permission position, the domains and subdomains you know of before starting, and the boundary statement in your own words.

Step 2 (60 min). Run the audit query set on Worksheet 3B. Minimum ten query patterns:

#	Query pattern	Looking for
1	Site restriction alone	Approximate indexed scale
2	Site restriction per known subdomain	Which are indexed
3	Site restriction with a wildcard subdomain form, if supported	Subdomains you did not know about
4	Site restriction + each document file type in turn	Indexed documents by format
5	Site restriction + URL terms for staging, test, development, or archive	Non-production content
6	Site restriction + title terms characteristic of directory listings	Automatically generated index pages
7	Site restriction + an exact phrase that should appear on exactly one page	Duplicate content
8	Site restriction + a before-date restriction	Legacy content still indexed

#	Query pattern	Looking for
9	Site restriction + terms characteristic of internal document markings	Documents that may not have been meant for publication
10	Exact organization name + exclusion of their own domain	What third parties publish about them

Step 3 (20 min). Findings classification on Worksheet 3C: for each finding, the type (forgotten subdomain, unexpected document, duplicate, legacy content, possible unintended publication, third-party mention), the evidence, and the severity from the site owner's perspective.

Step 4 (15 min). Duplicate analysis on Worksheet 3D: for any content appearing at several URLs, list the URLs, assess whether it is genuine duplication or a technical artifact, and note the consequence for the site.

Step 5 (15 min). Boundary log on Worksheet 3E: record every point at which you decided not to do something, and why. **This is graded and it is the most important worksheet in the lab.**

Step 6 (20 min). Write the audit report: one page to a hypothetical site owner, with the indexed scale, the findings by type, a prioritized remediation list, and an explicit statement of what the audit did and did not do.

Worksheet 3A: Scope and Permission

Element	Value
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Worksheet 3B: Audit Queries

#	Query pattern used	Result count	Findings	Notes
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Worksheet 3C: Findings

Finding	Type	Evidence	Severity to the owner
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Worksheet 3D: Duplicates

Content	URLs	Genuine duplication?	Consequence
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Worksheet 3E: Boundary Log

Point at which I stopped	What I could have done	Why I did not
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Expected Outcomes

- An appropriate subject, with the permission position recorded honestly.
- **Ten-plus query patterns executed**, with counts and findings.
- **The indexed document count should substantially exceed the student's prior expectation.** On most institutional sites it does, frequently by a factor of three or more, and noticing that is the lab's headline finding.
- **At least one subdomain the student did not know about**, on most institutional subjects.
- **At least one duplicate or near-duplicate**, with the URLs recorded.
- **A boundary log with genuine entries.** A student who records nothing in Worksheet 3E either found nothing interesting or did not notice the moments where they could have gone further. The strongest submissions record two or three specific points of restraint.
- **An audit report that states what was not done**, which is the professional shape of this deliverable.
- If anything resembling an accidental exposure was found: recorded, not accessed, and handled per Chapter 7.

Discussion Questions

1. How many indexed documents did you expect, and how many were there?
2. Which subdomain surprised you?
3. Where did you stop, and what made that the right call?
4. If this were your organization, which finding would you remediate first?
5. Why does the boundary log matter more than the findings list?

Grading Rubric: Lab 3 (60 points)

Criterion	Weight	4 points looks like
Subject appropriateness and permission	x3	Appropriate choice, position recorded honestly
Boundary statement in own words	x2	Accurate and specific
Query coverage	x3	Ten-plus patterns with counts
Subdomain enumeration	x3	Known and discovered subdomains, with at least one unexpected
Document inventory	x3	Per format, with the count against expectation
Duplicate detection	x2	Identified with URLs and

Criterion	Weight	4 points looks like
		consequence
Findings classification	x3	Typed and severity-rated from the owner's perspective
Boundary log	x4	Genuine restraint points recorded with reasoning
Exposure handling, if applicable	x2	Recorded, not accessed, Chapter 7 applied
Audit report	x3	Prioritized, and states what was not done

Total possible: 112 weighted. Passing: 77.

Lab 4: Structured OSINT Research on an Organization

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate to advanced

Objectives

1. Run a structured research sequence using advanced operators against an organizational subject.
2. Document every query, its purpose, and its yield.
3. Evaluate source credibility for what the queries return.
4. Refine the strategy in response to results, and record the refinements.

Prerequisites

- All of Part 1, and Chapters 1, 3, and 9 for source evaluation.
- Labs 1 through 3 complete.

Choose a Track

Track	Subject	Why
A: Live	A real public institutional organization: a public agency, a university, a port or transit authority, a utility, or a large registered charity. Not a private individual, not a small business, not your employer's competitor without authorization	The operators must actually be executed to be learned
B: Provided packet	The fictional organization in	For anyone who cannot search

Track	Subject	Why
	Section 8.4.1, using the simulated results	live

The client frame is fictional in both tracks. You are researching on behalf of **Calloway Grants**, a fictional funder conducting due diligence before a grant decision. Your questions are the same either way:

#	Question
Q1	What does the organization publish about its own structure, governance, and locations?
Q2	What documents has it published, of what types, and over what period?
Q3	What do third parties publish about it?
Q4	What official or regulatory material exists about it?
Q5	What has it published previously that it no longer publishes?

8.4.1 Provided Packet (Track B)

*Fictional. Simulated results for **Fenwick Coastal Trust**, an invented environmental charity. Each row is a result, with the query pattern that produced it.*

#	Query pattern	Result	Type
1	Site restriction alone	214 indexed pages	Count
2	Site restriction + PDF	46 documents: 9 annual reports, 14 project reports, 8 policy documents, 15 newsletters	Documents
3	Site restriction + spreadsheet formats	3 files: two grant schedules, one volunteer rota	Documents
4	Site restriction + presentation format	5 conference decks, most recent 3 years old	Documents
5	Site restriction + <code>intitle: governance terms</code>	Trustee page listing 9 trustees; a governance policy page	Pages
6	Site restriction + <code>inurl: for a projects</code>	31 project pages across four regions	Pages

#	Query pattern	Result	Type
	path		
7	Exact organization name + exclusion of their domain	87 results: local press, two funder announcements, a regulator entry, several directory listings	Third party
8	Exact name + regulator domain suffix	The charity regulator's register entry, with filing history	Official
9	Exact name + filetype:pdf + exclusion of their domain	Four documents on other organizations' sites naming Fenwick as a partner	Third party documents
10	Exact name + a date window covering five years ago	Coverage of a merger with a smaller trust	Third party
11	Site restriction + before: five years ago	Pages describing two projects that no longer appear on the current site	Legacy
12	Site restriction + intitle: for annual report + numeric range across years	Annual reports for six of nine years; three years missing from the indexed set	Documents
13	Exact name AROUND a regional term	Eleven results tying the organization to a specific region, of which three are substantive	Third party
14	Exact name + terms characteristic of complaints or disputes	Two results: a planning objection and a forum thread, neither substantive	Third party
15	Site restriction + intitle: for financial terms	A finance policy page; no financial statements outside the annual reports	Pages

Additional context. The regulator entry (item 8) lists nine filings; the site publishes six annual reports (item 12). The merger (item 10) is not mentioned anywhere on the current site. Two of the four partner documents (item 9) describe projects not listed on the current projects pages.

Steps

Step 1 (20 min). Research plan on Worksheet 4A. For each of Q1 to Q5: the query patterns you will use, the operators in each, the reason for each operator, and what result type you expect. Minimum twelve planned queries.

Step 2 (65 min). Execute (Track A) or analyse the packet (Track B), recording every query on Worksheet 4B: exact query or pattern, purpose, result count, yield, and the findings by item.

Step 3 (25 min). Source credibility assessment on Worksheet 4C. For every finding: the source class (subject's own publication, regulator or official, press, partner or third-party document, community, archive), reliability and credibility ratings per Chapter 1, and whether the finding is corroborated elsewhere.

Step 4 (25 min). Discrepancy analysis on Worksheet 4D. Identify every point where sources disagree or where the subject's own material is incomplete. State what each discrepancy establishes and what it does not.

Step 5 (20 min). Refinement record on Worksheet 4E: every query you changed and why, distinguishing operator refinements from vocabulary refinements, and the effect of each on count and yield.

Step 6 (25 min). Write the research summary: one page for the fictional funder, answering Q1 to Q5, with facts separated from assessments, every fact attributed to its source class, confidence levels, ranked gaps, and an explicit statement of what search operators could not establish and which structured source would.

Worksheet 4A: Research Plan

Question	Query pattern	Operators and reasons	Expected result type
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Worksheet 4B: Execution Log

#	Query	Purpose	Count	Yield	Findings
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Worksheet 4C: Source Credibility

Finding	Source class	Reliability	Credibility	Corroborated?
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Worksheet 4D: Discrepancies

Discrepancy	Sources	What it establishes	What it does not
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Worksheet 4E: Refinements

Original query	Refined query	Operator or vocabulary?	Effect on count	Effect on yield
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Expected Outcomes

- Twelve-plus planned queries with per-operator reasons, produced before execution.
- **The filing discrepancy is the central analytic finding in Track B:** the regulator lists nine filings and the site publishes six annual reports. That establishes that three years of reports are not on the site or not indexed, and it does not establish that the organization failed to file, because the regulator's record shows it did. Students who report this as a compliance failure have over-read it.
- **The merger is absent from the subject's own material** and appears only in third-party coverage from five years ago, which is a Q5 finding and a demonstration of why the exclusion query at item 7 matters.
- **Two partner documents describe projects not on the current site**, which combined with the legacy pages at item 11 indicates that the organization's published history is incomplete. This is an observation about their publishing, not about their activity.
- **Source class discipline:** the subject's own publications are primary for what the subject says and are not independent; the regulator entry is the highest-authority item in the packet; the forum thread is the lowest.
- **The Q4 answer should route to the regulator's own system**, not to search results, per Chapter 3 and Chapter 9.
- **The summary must state what operators could not establish**, which in Track B includes finances beyond the published reports, the reason three reports are missing, and anything about the merger's terms.
- Refinement record showing at least two vocabulary refinements alongside operator refinements.

Discussion Questions

1. The regulator lists nine filings and the site publishes six. Write the sentence you would put in the summary.
2. The merger appears nowhere on the subject's own site. What does that establish about the subject?
3. Which of your five questions could not be answered with operators at all?
4. Which source class did you rate highest, and would that hold for a different kind of subject?
5. What would you do next, and would it involve a search engine?

Grading Rubric: Lab 4 (64 points)

Criterion	Weight	4 points looks like
Research plan before execution	x3	Twelve-plus queries with per-operator reasons
Execution log completeness	x3	Every query, count, yield, findings
Coverage of all five questions	x3	Each addressed with appropriate operators
Source class assignment	x3	Correct classes; subject's own material identified as non-independent
Reliability and credibility ratings	x3	Chapter 1 framework applied consistently
Filing discrepancy handling	x4	Identified and correctly limited, not over-read
Legacy and absence findings	x3	Merger absence and legacy pages identified as publishing observations
Discrepancy analysis	x3	Establishes and does not establish, per discrepancy
Refinement record	x2	Operator and vocabulary refinements distinguished
Summary quality	x3	Facts separated, attributed, confidence stated, gaps ranked
Structured source routing	x2	Q4 routed to the regulator's system rather than to results

Total possible: 128 weighted. Passing: 88.

Lab 5: Troubleshooting Broken Queries

Time required: 2 to 2.5 hours **Format:** Individual, then pair comparison **Difficulty:** Intermediate

Objectives

1. Diagnose syntax and structural errors in queries.
2. Repair each and predict the effect of the repair.
3. Compare results before and after.
4. Build a diagnostic routine.

Prerequisites

- Sections 1.2 through 1.8, and comparison table 2.8.
- Lab 1 complete; bring the reference card.

The Broken Query Set

Fifteen queries, each with at least one problem. Some have several. Two of them are syntactically correct and fail for a different reason, and identifying those is part of the exercise.

#	Query as written	Intended purpose
1	site: gov.uk flood risk assessment	Government flood risk documents
2	"annual report" filetype:.pdf site:example.org	An organization's annual reports
3	climate adaptation or resilience funding	Either term, with funding
4	allintitle:coastal erosion site:ac.uk filetype:pdf	Academic PDFs titled about coastal erosion
5	drainage - highway maintenance	Drainage, excluding highway maintenance
6	site:example.gov site:example.org budget	Budgets from either domain
7	filetype:xls filetype:csv rainfall data	Rainfall data in either format
8	cache:example.org/reports	The stored copy of a reports page
9	"flood defence" around(10) "cost benefit"	The two phrases near each other
10	manufact* standards	Manufacturing, manufacturer, manufactured
11	intitle:"strategy" before:2020	Strategy documents before 2020
12	"site:example.org annual report 2023"	An organization's 2023 annual report
13	link:example.org	Pages linking to the organization
14	"coastal management plan" filetype:pdf	Recent coastal plans from one domain

#	Query as written	Intended purpose
	site:example.gov after:2024-01-01	
15	~flooding mitigation techniques	Flooding and its synonyms, with mitigation

Steps

Step 1 (35 min). Diagnosis on Worksheet 5A. For each query: every error present, the error type from table 2.8, what actually happens when it is run, and what the user probably expected.

Step 2 (30 min). Repair on Worksheet 5B. Write the corrected query for each, and **predict** the effect of the repair on the result count and on relevance before running anything.

Step 3 (30 min). Execution and comparison on Worksheet 5C. Run both the broken and the repaired version of each query where it is safe and sensible to do so. Record both counts, the yield in the first ten results for each, and whether your prediction held.

Step 4 (15 min). The two special cases. Two of the fifteen are syntactically correct and still fail. Identify them, explain why, and state what the user should do instead. **These are graded heavily.**

Step 5 (20 min). Diagnostic routine. Write a repeatable routine, in no more than eight steps, that you would follow when any query behaves unexpectedly. It must handle both the zero-results case and the too-many-results case.

Step 6 (15 min, in pairs). Exchange routines with a partner and test each other's against three of the broken queries. Record where the routine failed to catch an error.

Worksheet 5A: Diagnosis

#	Errors present	Error type	What actually happens	What the user expected
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Worksheet 5B: Repair and Prediction

#	Repaired query	Predicted count change	Predicted relevance change
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Worksheet 5C: Comparison

#	Broken count	Repaired count	Broken yield	Repaired yield	Prediction held?
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Expected Outcomes

The diagnoses should include:

#	Errors
1	Space after the operator colon, so the site restriction is ignored and gov.uk becomes a search term
2	Dot in the file type value
3	Lowercase or, and ungrouped scope even if capitalized
4	allintitle: combined with other operators, which will suppress them; also the domain suffix form should be tested
5	Space after the minus, so it is treated as text
6	Two site restrictions, which conflict; needs grouped OR
7	Two file type restrictions, which conflict; needs grouped OR
8	Deprecated operator: the cache operator has been withdrawn
9	Lowercase around(10); also the interaction with quoted phrases is unreliable
10	Asterisk used as a stem wildcard, which does not work; needs explicit OR of the variants
11	Syntactically correct. It will run. It fails because a single-word title restriction on a generic word plus a broad date bound is far too loose to be useful, and because the inferred date is unreliable. The repair is a more specific title phrase and a bounded window
12	Operators inside the quotation marks, so the whole string is searched literally
13	Deprecated operator: the link operator has been withdrawn
14	Syntactically correct. It will run and may return nothing, because the constraints are legitimately narrow. The correct response is not to repair the syntax but to relax one constraint at a time, per the newest-constraint rule, and to record the negative properly if it persists
15	Tilde operator, withdrawn; needs explicit OR of

#	Errors
	chosen synonyms

- **Items 11 and 14 are the two special cases**, and identifying them as correct-but-failing rather than as syntax errors is the lab's central test. Item 14 in particular teaches that a null result from a correct query is a coverage finding, not a bug, which is Rule 10.
- Predictions recorded before execution, with at least some proving wrong.
- A diagnostic routine of eight steps or fewer that handles both failure directions.

Discussion Questions

1. Which error would you be most likely to make yourself under time pressure?
2. Items 11 and 14 are syntactically correct. What does that tell you about diagnosing queries by syntax alone?
3. Three of the fifteen use withdrawn operators. How would you have known without this chapter?
4. Where did your prediction about the repair's effect prove wrong?
5. Your partner's routine caught something yours missed. What was it?

Grading Rubric: Lab 5 (56 points)

Criterion	Weight	4 points looks like
Error identification	x4	All errors in all fifteen, correctly typed
Multiple errors per query caught	x2	Where several exist, all found
Deprecated operators identified	x3	All three, with alternatives named
Repairs correct	x3	Every repair works and preserves the intent
Predictions recorded before execution	x3	Written first, compared honestly
Comparison data	x2	Counts and yields for both versions
Items 11 and 14 identified as correct-but-failing	x4	Both, with the correct explanation and response
Rule 10 applied to item 14	x3	Null result treated as a coverage finding
Diagnostic routine	x3	Eight steps or fewer, handles both directions
Pair testing record	x1	Gaps found and recorded

Total possible: 112 weighted. Passing: 77.

Part 9: Capstone Assignment

The Harrowgate Estuary Crossing Research Brief

Weight: 35% of chapter grade **Time required:** 5 to 8 hours **Format:** Individual **Deliverables:** Six items, specified in Section 9.3

9.1 The Scenario

The client and the brief are fictional. The research is real, because the subject is a real category of public infrastructure.

You are engaged by **Ravensworth Policy Associates**, a fictional research consultancy, to produce a research brief for a client considering a submission to a public consultation on a **major estuary or river crossing project**.

Choose a real, publicly documented major crossing project in any country: a bridge, a tunnel, or a combined scheme, ideally one with a public consultation history. It must be a public infrastructure project with substantial published material. Nothing in this assignment targets a private individual or a private company's confidential affairs.

The client's five questions.

#	Question
Q1	What official documentation exists for the project: consultation documents, environmental assessments, business cases, and decision records?
Q2	What is the project's documented timeline, including changes to scope, cost, or schedule?
Q3	What technical and academic material exists about crossings of this type?
Q4	What have third parties published: press coverage, objections, supporting submissions, and analysis?
Q5	What material existed previously and no longer appears in the current published record?

Constraints. Five working days of notional effort. Search operators are your primary tool, and the brief must be honest about their limits.

9.2 Guidance on the Substance

Q5 is the question operators alone cannot answer, and the strongest submissions recognize that early and route it to archives, per Chapter 3.

Q1 will largely resolve to structured sources: a scheme's official document library, a planning inspectorate or equivalent, and a transport authority's publication system. The operators' job is to find those systems and then to find specific documents inside them where the systems' own search is poor, which case study 3.2 established is common.

Progressive refinement is the graded behaviour. The capstone asks for the sequence, not just the final queries. A submission showing twelve well-chosen final queries with no record of how they were reached will score below one showing thirty queries in a visible progression from broad to precise.

9.3 What You Must Produce

Deliverable 1: Search Strategy (16 points)

- The five questions decomposed into searchable propositions, minimum fifteen.
- For each proposition: the query pattern intended, the operators, and the reason for each operator choice.
- A vocabulary harvest: minimum ten terms taken from real documents, with the source of each, and a note on which of your initial terms they replaced.
- The name and entity variant list for the project, its promoter, and its regulator, minimum eight variants including any former scheme names.
- An explicit statement of which questions you expect operators to answer and which you expect to route elsewhere.

Deliverable 2: Progressive Query Log (20 points)

Minimum thirty logged queries, showing progression. Every entry: query number, exact query string, the operators used and whether each worked, purpose, locale and language settings, signed-in state, date and time with time zone, reported result count, yield in the first ten, findings by item, and the refinement made next with its reason.

The log must show at least four documented refinement chains, meaning a sequence of three or more queries in which each is a deliberate modification of the last, with the reason recorded.

Deliverable 3: Findings by Question (20 points)

For each of Q1 to Q5:

- The findings, each with the query that produced it, the source class, and reliability and credibility ratings.

- Which findings came from a structured source that a query identified, rather than from the query itself.
- Conflicts between sources, recorded and not resolved by preference.
- A confidence level per question with the reason.
- For Q5 specifically: the archive work, with the capture dates and what changed.

Deliverable 4: Operator Contribution Analysis (14 points)

- For each operator you used: how many queries used it, what it contributed, and one specific finding it was responsible for.
- The two operators that contributed most, with evidence.
- Any operator that failed, was ignored, or behaved unexpectedly, with what you observed.
- **A count of improvements attributable to vocabulary rather than operators**, with examples.
- What you would have missed with no operators at all.

Deliverable 5: Research Brief (24 points)

Maximum four pages plus annexes. Addressed to the fictional client.

Required: a summary of no more than four sentences; answers to Q1 to Q5 with facts separated from assessments and every fact attributed to a source; a documented timeline for Q2 (Chapter 5 conventions, with precision markers); **a limitations section treating search limitations as substantive content**, covering index dependence, result depth, inferred dates, personalization, and what operators could not reach; an ethical considerations section covering Rule 11, what you did not pursue and why, and the treatment of any individual who appeared in the material; ranked gaps each with the source that would close it; and a methodology refinement section stating what you would do differently. Annexes: the query log, the operator contribution analysis, and a source inventory.

Deliverable 6: Reflection (6 points)

Two pages maximum: which operator you understood least well before this chapter; where a query failed silently and how you noticed; the largest single improvement in your search practice; and one specific habit you will keep.

9.4 Capstone Grading Rubric (110 points, scored to 100)

Deliverable	Criterion	Points
1. Strategy	Fifteen-plus searchable propositions	4
	Per-operator reasoning	4
	Vocabulary harvest, ten-plus	4

Deliverable	Criterion	Points
	terms with sources	
	Entity and variant list, eight-plus	2
	Operator-answerable versus route-elsewhere statement	2
2. Log	Thirty-plus entries to the full standard	8
	Four-plus documented refinement chains	6
	Operator behaviour recorded, including failures	3
	Negatives with coverage boundaries	3
3. Findings	All five questions addressed	4
	Source class and ratings per finding	4
	Structured-source findings distinguished from query findings	4
	Conflicts recorded, not resolved by preference	3
	Q5 archive work with capture dates	5
4. Operator analysis	Per-operator contribution with a specific finding each	5
	Top two identified with evidence	3
	Failures and anomalies recorded	3
	Vocabulary versus operator improvements counted	3
5. Brief	Summary: four sentences, judgment, confidence	3
	Five questions answered, facts attributed and separated	5
	Timeline with precision markers	3
	Limitations as substantive content, not a caveat	5
	Ethical considerations including Rule 11	4
	Ranked gaps with resolving	2

Deliverable	Criterion	Points
	sources	
	Methodology refinement	2
6. Reflection	Specific, honest, with a retained habit	6
Total		110, scored to 100

Grade bands: 90 to 100 exemplary, 80 to 89 proficient, 70 to 79 competent, 60 to 69 developing, below 60 not yet passing.

Automatic deductions:

- Any attempt to access material beyond what search results and publicly served pages display, or any use of operators against systems you are not authorized to examine: **minus 30**. Rule 11.
- Any automated querying: minus 20. Rule 5.
- Researching a private individual, or including one beyond an official capacity: minus 20.
- Stating that material does not exist on the basis of null search results: minus 12. Rule 10.
- A query log below twenty entries: capped at 70; below ten: capped at 55.
- No refinement chains, meaning the log shows only final queries: minus 10.
- Treating search results as findings without going to the structured sources they identify: minus 10.
- No limitations section, or a limitations section that is a single disclaimer sentence: minus 8.
- Citing a reported result count as a finding: minus 5 per instance.

Guidance on the expected shape of a strong package. The strategy should show that Q1 routes mostly to structured document libraries which the operators locate, that Q3 routes to scholarly and grey literature via domain suffix and file type patterns, and that Q5 routes to archives. The query log should be visibly iterative, with early broad queries harvesting vocabulary that later queries use, and with at least one chain that ends in a well-documented negative. The operator contribution analysis should almost certainly identify the site restriction and the file type restriction as the top two, because they are, and should honestly report that several improvements came from better terms rather than better operators. The limitations section is worth five points because it is the part students treat as boilerplate and it is where professional judgment actually shows: index dependence, unreachable result depth, inferred dates that must not be cited as publication dates, personalization you cannot see around, and the categories of material that no query reaches. And the ethical section should record something specific that was not pursued, because on any large infrastructure project the material will include individuals, objectors, and correspondence, and the discipline of leaving those alone is the point.

Part 10: Answer Keys

Instructors may wish to remove this section from student copies. Operator behaviour changes; verify before teaching from these answers.

10.1 Short Answer Key

1. An operator constrains or modifies matching and ordering within the existing index. It cannot expand the index because it acts on what has already been crawled and stored; a page never fetched, never indexed, or since dropped cannot be returned by any query formulation.
2. Precision is the proportion of returned results that are relevant; recall is the proportion of relevant material returned. The working pattern is high recall first, to see the shape of the material and learn its vocabulary, then rising precision to work it.
3. Documented operators carry a stated support commitment and are more stable. Undocumented operators are supported at Google's discretion, can change or vanish without notice, and their behaviour is inferred from observation, sometimes wrongly. Both can fail; the undocumented ones fail without announcement.
4. Choose a query with a known answer, run it without the operator and record the count and a known result, run it with the operator and record the same, compare, then construct a query that must return nothing if the operator is working. The negative test is essential because an ignored operator returns plausible results that look filtered, and only a must-return-nothing query exposes it.
5. Silent failure is an operator being accepted and ignored. Detection: construct a query that could only return results if the operator is not working, such as a file type restriction to an extremely rare format combined with a common term; if millions of ordinary pages return, the operator is being ignored.
6. AND is implicit between terms. OR must be capitalized or written as a pipe. NOT is an unspaced minus before a term. The most common error is lowercase `or`, which is treated as an ordinary word and fails silently, returning an implicit AND result that looks plausible.
7. Because distinctive strings are the highest-yield queries and quotation marks preserve them exactly, while also suppressing synonym expansion and stemming. Use cases: organization and document names, error messages and technical strings, boilerplate and template language for provenance tracing, and quoted sentences for finding where a claim first appeared.
8. Exclusion applies to phrases, domains, file types, titles, and URL patterns, not only single words. The risk is that over-exclusion removes relevant material invisibly, so exclusions should be added one at a time and removed one at a time when a query returns too little.
9. Enumerating an organization's own published material; sweeping a sector by domain suffix to find institutions you did not know existed; auditing what is indexed against what an organization believes it publishes; and excluding a dominant domain to find third-party coverage.
10. A page's title is the closest thing to a machine-readable statement of what the page is about, while a body term may be incidental. Moving a term from a body requirement to a title requirement is the fastest single precision improvement available.

11. The `all` prefixed operators tend to claim the whole query and suppress other operators. Use the `in` prefixed forms repeatedly instead, which combine freely.
12. A site restriction or domain suffix, plus a file type restriction, plus an exact phrase from the expected title or content. It is the most productive shape because documents rather than pages carry investigative substance, and this combination targets the artifact on the right domain with the right title.
13. Because freshness bias buries original reporting under recent restatement, and date restriction inverts that. The caution is that the date is inferred and may be a publication, modification, extracted, or crawl date, so it is a strong filter and a weak fact and must never be cited as a publication date.
14. Two terms on the same page may be unrelated, particularly on long pages and archives; two terms within a few words are in the same thought. The proximity operator addresses this, with the caveats that it is undocumented, capitalized, and approximate.
15. Any five of: the plus prefix (use quotation marks), the tilde prefix (use explicit OR), `link:` (use backlink services), `info:` (visit the page), `cache:` (use web archives), `related:` (use directories and association lists), `daterange:` (use before and after), directory-lookup operators (nothing, and Rule 1 prohibits the use case).
16. Because investigators used it daily to see a page's prior state, and its withdrawal removed that capability from the engine entirely. Web archives fill the gap and are better for the purpose: dated, independent of the engine, and preserving multiple states.
17. It is good for noticing that a query changed by an order of magnitude when an operator was added, which is a diagnostic that the operator did something. It is not good for measuring how much material exists, comparing indexes, or citing in a report, because it is approximate, unstable between refreshes, and mostly unreachable through paging.
18. Broad first, harvest vocabulary, add structure, observe the effect, narrow one constraint at a time, and record everything. When results collapse to zero, remove the most recently added constraint, because the failure is almost always the newest addition rather than the query as a whole.
19. Ownership questions go to property records; litigation to court systems; registration to corporate registries; permits to permit portals; prior page states to archives; peer-reviewed evidence to scholarly databases. The operator's job in those cases is to find out which system holds the record.
20. Rule 11: an operator that finds something does not authorize accessing it. Legitimate uses are auditing your own footprint, auditing a client's with written authorization, and understanding the technique defensively. Illegitimate uses are searching for exposed systems you do not own, accessing anything surfaced on a third party's system, collecting exposed personal data, and testing credentials.

10.2 Multiple Choice Key

#	Answer	Note
1	b	Filters and reorders the existing index

#	Answer	Note
2	b	No space after the colon; phrase quoted
3	b	Lowercase or is an ordinary word
4	c	Title restriction
5	b	The a l l prefix problem
6	b	The document discovery pattern
7	b	An inferred date
8	b	Approximate distance, capitalized
9	c	The cache operator
10	c	Web archives
11	b	An order-of-magnitude diagnostic
12	b	The newest-constraint rule
13	b	Conflicting conditions; use grouped OR
14	c	Exposure by error is not publication
15	b	What the team did not do

10.3 True or False Key

#	Answer	If false, why
1	False	No operator can return a page that was never crawled or indexed
2	False	OR and wildcards increase recall; not all operators narrow
3	False	Lowercase or is treated as an ordinary word and fails silently
4	True	
5	False	The a l l prefixed operators combine poorly and frequently suppress other operators
6	False	The date is inferred and may be a modification, extracted, or crawl

#	Answer	If false, why
		date
7	False	It is an unstable approximation and most of the claimed results are unreachable
8	False	Several have been withdrawn, including the cache, link, info, plus, and tilde operators
9	True	
10	False	Exposure by error is not publication; finding it does not authorize accessing it

10.4 Critical Thinking Guidance

7.1 (a) Site restriction on the university domain, plus file type, plus a title phrase. (b) Exact phrase, unrestricted, then the same phrase with the known site excluded. (c) Exact model or product string, plus file type, plus a date window covering the release period. (d) Exact place and decision terms, plus a local press domain restriction or the news location operator. (e) Site restriction plus a before-date restriction, then archives for what has been removed. (f) Site restriction plus each file type in turn, per Lab 3.

7.2 Three examples: what the manufacturer publishes (site restriction plus file type); what third parties say (exact name plus exclusion of their domain); and whether they are registered and in good standing (route to the corporate registry, not to an operator). Each excludes the others' material by design.

7.3 Grouped OR across data file types, plus a subject phrase, plus a domain suffix likely to hold the data. An unrestricted query fails because pages about data outrank data files, and a single file type restriction misses the other formats.

7.4 (a) Space after the colon; remove it. (b) Dot in the file type; remove it. (c) Lowercase `or` and ungrouped scope; capitalize and group. (d) `allintitle:` with other operators; use `intitle:` twice. (e) Space after the minus; remove it. (f) Two site restrictions; use grouped OR.

7.5 Likely problems: no exact phrases, no structural operators, and generic vocabulary. Three changes in order: quote the distinctive phrase; add a site restriction or a file type; and replace generic terms with terms harvested from a real document.

7.6 The response should cover Rule 10, the newest-constraint rule, and a diagnostic sequence: remove the last operator, then quote or unquote phrases, then check for syntax errors, then broaden vocabulary, then try a second engine, and only then record a negative with its coverage boundary.

7.7 The query combines a file type, a domain suffix, a title phrase, a date window, and a domain exclusion. What could go wrong: the date restriction may exclude undated documents; the title phrase may be too specific; the domain suffix may not be supported as written. Test each by removing it and

observing the count.

7.8 A grouped OR of the four names, a grouped OR of the two file types, and no domain restriction. Both groups must be parenthesized, or the scope of each OR becomes ambiguous.

7.9 Two file type restrictions conflict, because a document cannot be two formats, so the conditions cannot both be satisfied; use a grouped OR. The same principle applies to multiple site restrictions and to multiple language restrictions.

7.10 Different locales; different personalization or signed-in state; different times, because indexes change; and different devices. Checks: compare locale settings, search signed out in a clean profile, note the timestamps, and compare on the same device type.

7.11 It tells you that of 12,000 pages mentioning the term, only six are about it. That is a good outcome if you wanted documents on the subject, and a bad one if you wanted every mention, for example for provenance tracing.

7.12 Either the operator is being ignored, or the operator's constraint was already satisfied by the rest of the query. Distinguish by constructing a negative test that must return nothing if the operator is working.

7.13 Graded on whether the student defines a stopping condition in advance rather than searching until they feel finished. Chapter 4's saturation concept is the reference.

7.14 The negative should name the exact queries, operators, locale, dates, and engine, and state what it does not establish. The three things to try: different vocabulary from a real document, a different engine with an independent index, and a structured source that would hold the material if it exists.

7.15 Graded on the honesty of the prediction comparison rather than on the query.

Appendix A: Reusable Templates

A.1 Query Log Entry

Field	Value
Query number	
Exact query string	
Operators used	
Each operator appeared to work?	
Purpose	
Locale and language	
Signed-in state and profile	
Date, time, time zone	

Field	Value
Reported result count	
Yield in the first ten	
Findings by item ID	
Refinement made next, and why	
Screenshot reference	

A.2 Operator Test Record

OPERATOR: _____ SYNTAX AS ENTERED: _____
 PURPOSE: _____
 PREDICTED BEHAVIOUR: _____
 BASELINE QUERY: _____ COUNT: _____
 Known result present? Y / N
 WITH OPERATOR: _____ COUNT: _____
 Known result present? Y / N
 NEGATIVE TEST (must return nothing if working): _____
 RESULT: _____
 VERDICT: works / partially / ignored / withdrawn
 LIMITATIONS OBSERVED: _____
 TESTED ON (date): _____ BY: _____

A.3 Query Design Sheet

Element	Content
Question	
What artifact would answer it	
Vocabulary harvested from real documents	
Operators chosen, with a reason each	
Predicted result type	
Predicted count magnitude	
Fallback if it returns nothing	
Fallback if it returns too much	

A.4 Refinement Chain Record

Step	Query	Count	Yield	Change made	Reason	Effect
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A.5 Website Audit Query Set

#	Pattern	Looking for	Count	Findings
1	Site restriction alone	Indexed scale		
2	Site restriction per subdomain	Which are indexed		
3	Site restriction + each file type	Indexed documents		
4	Site restriction + URL terms for non-production paths	Staging and test content		
5	Site restriction + title terms for directory listings	Generated index pages		
6	Site restriction + a should-be-unique phrase	Duplicate content		
7	Site restriction + before-date	Legacy content		
8	Site restriction + internal marking terms	Possible unintended publication		
9	Organization name + exclusion of own domain	Third-party mentions		
10	Site restriction + governance or contact title terms	Structural pages		

A.6 Diagnostic Routine

QUERY RETURNS NOTHING

1. Remove the most recently added operator. Recheck.
2. Check for spaces after colons and minus signs.
3. Check quotation marks are straight, not typographic.
4. Check OR is capitalized and grouped.
5. Check for conflicting operators (two site or two file type restrictions).
6. Replace the phrase with a shorter, more common one.
7. Remove the date restriction.
8. If it still returns nothing: this may be a genuine coverage finding. Record it with its boundary.

QUERY RETURNS TOO MUCH

1. Quote the distinctive phrase.
2. Add a site restriction or a domain suffix.
3. Add a file type restriction.
4. Move a key term into a title restriction.
5. Exclude the dominant irrelevant domain or term.
6. Add a date window.
7. If it is still noisy: the problem is vocabulary.
Harvest better terms from a real document.

A.7 Negative Finding Template (operator-specific)

Engine: _____ Locale: _____ Language: _____
Exact query: _____
Operators used: _____
Each operator verified as working? _____
Date/time (TZ): _____
Result: no relevant results

THIS DOES NOT ESTABLISH that the material does not exist.
It establishes that no page matching these terms, under
these constraints, is indexed by this engine as of this
date.

VARIANTS TRIED: _____
OPERATORS RELAXED IN TESTING: _____
OTHER ENGINES TRIED: _____
STRUCTURED SOURCE THAT WOULD SETTLE IT: _____

Appendix B: The Operator Quick Reference

Verify before relying. Test date: _____

Purpose	Operator	Form
Exact phrase	quotation marks	"exact phrase"
Exclude	minus, unspaced	-term
Either	OR, capitalized, or pipe	a OR b
Group	parentheses	(a OR b) c
Any word in a slot	asterisk	"a * b"
Number range	two dots	2015..2020
One domain, subdomain, path, or suffix	site	site:example.org
Exclude a domain	minus plus site	-site:example.org
Term in the URL	inurl	inurl:reports
Term in the title	intitle	intitle:"annual report"
Term in the body	intext	intext:"registration"

Purpose	Operator	Form
Document format	filetype	filetype:pdf
Several formats	grouped OR	(filetype:xls OR filetype:csv)
Before a date	before	before:2020-01-01
After a date	after	after:2018-01-01
Words near each other	AROUND, capitalized	a AROUND(5) b
Definitions	define	define:term

Withdrawn or degraded, do not rely on: cache:, link:, info:, related:, daterange:, +term, ~term, inanchor:, directory-lookup operators.

The two to learn first: site: and filetype:. They appear in six of the seven case studies in this chapter.

Appendix C: Where to Go Next

Skills to build after this chapter, in priority order:

1. **Test your operators quarterly** using Lab 1's procedure, and date the reference card. Operators are withdrawn quietly, and a stale reference is worse than none.
2. **Practise vocabulary harvesting** until it is automatic. For a month, run one broad query before every structured one, and note the terms you would not have used.
3. **Build the negative test habit.** For every operator you depend on, keep a query that must return nothing if it is working.
4. **Audit your own organization's indexed footprint** annually, per Lab 3, and report it to whoever owns the site.
5. **Learn the structured sources for your field** well enough that you stop reaching for an operator when a registry would answer.
6. **Practise provenance tracing** with exact phrases and date bracketing until finding the earliest indexed appearance is routine.
7. **Keep a personal query pattern library**, in structural form, of the shapes that work for your recurring questions.

Habits to start now:

- Quote every phrase.
- Capitalize OR.

- Check for a space after the colon before blaming the index.
 - Test the operator before trusting it, and again in three months.
 - When a query is noisy, fix the words before adding operators.
 - When results collapse, remove the last thing you added.
 - Record the query, the locale, and the time, every time.
 - Remember that finding something is not permission to open it.
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End of Chapter 10